Subsidize Rent or Subsidize Homeownership? Evidence from a Hong Kong Experiment

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Abstract

Governments often promote homeownership, but the effects of ownership subsidies may depend on institutional design. This paper studies the impact of Hong Kong's Tenants Purchase Scheme, a program that allowed 183,700 regularly income-tested public housing tenants to buy ownership rights but restricted resale and leasing. Leveraging its incomplete roll-out between 1998 and 2006, I find that the program significantly altered the household income and size distributions in treated housing estates, with large increases of households with incomes exceeding income limits. Evidence suggests that incumbent tenants often purchased units to avoid income tests and reconfigured co-residence among family members to obtain additional public housing units, with the unintended consequence being to exacerbate hardship for excluded low-income populations. The findings highlight how institutional details such as income test requirements and resale and leasing restrictions critically shape the impacts of housing assistance programs.

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1 Introduction

The choice between subsidizing rental or homeownership carries profound implications for the effectiveness of housing assistance, yet its full consequences remain inadequately understood. Some scholars document benefits of subsidized ownership, such as reduced housing misallocation (Wang 2011), greater access to credit (Wang 2012), lower crime (Disney et al. 2023), increased wealth accumulation (Sodini et al. 2023), and higher labor supply (Jacob and Ludwig 2012, Zhang 2025). Others warn that subsidized homeownership may inflate housing prices (Hilber and Turner 2014, Sommer and Sullivan 2018), constrain geographical mobility (Battu, Ma and Phimister 2008), and dampen entrepreneurship (Blanchflower and Oswald 2013, Bracke, Hilber and Silva 2018). However, few studies have examined how institutional details may mediate the behavioral and distributional impacts of a government's choice between subsidizing ownership vs. rent. Key policy features—including income thresholds, resale restrictions, and leasing rules—vary widely across programs and nations. Examples range from Singapore's subsidized ownership schemes and Hong Kong's extensive public rental system to the UK's Right-to-Buy policy and diverse social housing models throughout Europe and Asia. A deeper understanding of implications of design details is essential for guiding optimal housing assistance policy.

This paper sheds light on the role of institutional design by assessing the impacts of Hong Kong's Tenants Purchase Scheme (TPS), a large-scale subsidized sale of public rental housing implemented between 1998 and 2006. Under TPS, 183,700 sitting tenant households—who had been regularly subject to income tests—were offered the opportunity to purchase their rental units at discounted prices. As in many other subsidized ownership programs, purchasers obtained permanent occupancy rights and were no longer subject to ongoing means testing. Unusually, however, TPS imposed stringent restrictions on transfers: owners could not lease or resell their units (except to eligible family members) without paying a very substantial premium. This combination of relaxed income testing and tight transfer restrictions creates a distinctive setting in which to study how specific institutional features mediate the effects of public housing programs.

The effects of TPS are estimated using a dynamic difference-in-differences design that exploits the staggered and incomplete rollout of the scheme across public housing estates. In the main specification, both later-treated and never-treated estates serve as controls, and the efficient estimator of Borusyak, Jaravel and Spiess (2024) is implemented to address the improper weighting problems associated with conventional two-way fixed-effects estimators under staggered treatment adoption. Estate-level outcomes—such as population, household size, household income, user cost of housing, and commute times—are constructed from restricted-access 10% and 20% random repeated cross-sections of the Hong Kong Population Census. To mitigate concerns that estimates are driven by pre-existing differences, treated estates are compared to control estates with similar construction years and reweighted to have similar demographics. I also flexibly control for demographic-specific and geography-specific time trends.

The estimates robustly reveal a large and persistent increase in the share of households in treated estates with incomes that were more than twice the Public Rental Housing (PRH) income limit. The share of households in treated estates with incomes above 2X the income limit rose by 5 percentage points, from an initial level of 11.4 percent, over the two decades following the sales. Given that only 77 percent of units in treated estates had been sold by that time, these estimates imply even larger effects of TPS on the share of above-threshold households among occupants of purchased units.

Why did the share of households above the PRH income thresholds increase so sharply in treated estates? Such households would have been required to pay 1.5X rent as public renters, but were no longer be subject to regular income tests after purchasing the unit through TPS. Therefore, as formally explained in a simple conceptual framework, TPS has three potential impacts. First, TPS may alter individual labor supply, by removing an effective tax on labor income. Second, TPS may alter household-level residential sorting, by softening incentives for higher-earning households to leave treated estates. Third, TPS may affect co-residence decisions within extended family networks—for example, by inducing lower-income relatives to move out and obtain additional public housing.

To assess the empirical relevance of these channels, the impact of TPS on auxiliary outcomes are examined. First, TPS is found to have led to a substantial decline in household size. Average household size in treated estates fell by about 4 percent within a few years, and soon became 7 percent lower than control within two decades. Total population decreased eventually decreased by around 10 percent—approximately 75,000 residents. By contrast, the effects of TPS on average household income at the estate level are statistically insignificant.

Second, TPS is found to have induced very limited residential sorting at the household level. Mobility of household heads was low in both treated and control estates and did not respond detectably to the policy. Moreover, exceedingly few purchasing households obtained or exercised the right to transfer units except to eligible family members, consistent with the stringent transfer restrictions. At the same time, the share of extended-family households fell substantially, while the share of nuclear-family households increased. Taken together, these patterns suggest that the decline in household size was driven primarily by changes in household composition rather than by changes in which households lived in the estates.

Third, the increases in average individual incomes and employment rates in treated estates are found to be partly attributable to changes in estate-level demographic composition. The share of residents with higher education increased markedly, indicating that the demographic composition of residents shifted toward more educated—and therefore higher-earning—individuals. The increase in income is also unlikely to reflect pressure from mortgage obligations, as average housing expenditures declined in treated estates.

The evidence is strongly suggestive that TPS induced a substantial reconfiguration of co-residence choices within extended family networks. Higher-income family members, no longer constrained by income tests but still constrained by transfer restrictions, were encouraged to remain in the purchased units. Meanwhile, lower-income relatives of incumbent households were induced to re-enter the public housing waiting list and obtain additional units.

Finally, a rough back-of-the-envelope calculation illustrates how TPS may have reduced access to subsidized rental housing for low-income households. The calculation suggests that the resulting reduction in available PRH units may account for roughly 10 percent of Hong Kong's subdivided private-sector units in 2016. A key implication is that careful attention to institutional design is crucial when determining whether, and how, to offer subsidized ownership versus rental housing.

This paper contributes to a growing set studies that examine the effects of the subsidized sale of public housing, many of which finds positive impacts. Wang (2011, 2012) provides evidence that the subsidized sale of state employee housing in China reduced housing misallocation, raised private-sector prices, relaxed credit constraints, and increased self-employment. Disney et al. (2023) present quasi-experimental evidence that UK's Right-to-Buy housing reform reduced crime due to behavioral changes of the incumbent population. Sodini et al. (2023) show that the subsidized sale of municipal-owned buildings in Sweden caused beneficiaries to experience wealth increases and increased consumption owing to property price appreciation.[1] The Hong Kong setting differs from prior studies, since leasing and resale restrictions prevented the realization of the benefits of homeownership that were emphasized in these studies. The unique setting underscores how the aggregate impacts of subsidized ownership may critically depend on institutional context.[2]

Our results also relate to a broader literature on the trade-offs involved in subsidizing homeownership. A first strand focuses on the mortgage interest deduction (MID). Glaeser and Shapiro (2003) show that in the United States the MID disproportionately benefits higher-income households and is an ineffective tool for promoting homeownership. Hilber and Turner (2014) provide reduced-form evidence that its capitalization into house prices largely offsets its intended effect, and in tightly regulated markets even reduces homeownership attainment. Using a dynamic general equilibrium model, Sommer and Sullivan (2018) find that the MID raises house prices, lowers homeownership, increases debt, and reduces welfare. A second strand emphasizes broader external costs of (subsidized) owner-occupation: Battu, Ma and Phimister (2008) document that homeowners exhibit lower geographical mobility and are less likely to exit unemployment via distant moves; Fischel (2001) describes "homevoters" who protect asset values by supporting restrictive land-use regulations and exclusionary zoning, contributing to higher housing costs; and Blanchflower and Oswald (2013) and Bracke, Hilber and Silva (2018) show that homeownership and mortgage debt can discourage entrepreneurship by re-

[1] Disney and Luo (2017) provide theoretical results regarding the welfare effects of UK's Right-to-Buy program, which shares many similarities with Hong Kong's Tenants Purchase Scheme.

[2] A related literature documents the effects of housing assistance on labor supply and child outcomes (e.g., Jacob 2004; Kling, Ludwig and Katz 2005; Jacob and Ludwig 2012; Chyn 2018; Dijk 2019).

ducing business start-ups and increasing risk aversion, respectively.

The literature on the targeting of social assistance is also related. The tension between targeting and allocative efficiency was first identified in theoretical work by Akerlof (1978) and Nichols and Zeckhauser (1982). Recent empirical studies have studied this tension in different types of non-housing welfare programs (Deshpande and Li 2019; Finkelstein and Notowidigdo 2019; Lieber and Lockwood 2019). A growing literature also studies how mechanism design for the initial allocation of public housing affects allocative and targeting efficiency (Waldinger 2021; Lee, Kemp and Reina 2022; Naik and Thakral 2022; Thakral and Murra-Anton 2024; Thakral Forthcoming). To date, however, little attention has been devoted to understanding how ownership rights *after* the initial allocation of subsidized housing affect the efficiency of public housing programs. This paper advances a better understanding of the trade-offs involved in the design of housing assistance programs by measuring the extent to which institutional details such as regular income tests and transfer restrictions were important in determining its effects. My findings reveal that TPS eroded targeting efficiency, due to strategic co-residence decisions within extended family networks—a mechanism that is typically ignored in existing literature, but has particular salience in societies with strong kinship ties, such as Hong Kong.

This study is among the first to use a quasi-experimental design to document the effects of housing assistance in Hong Kong. Existing studies on Hong Kong's Tenants Purchase Scheme and, more broadly, on Hong Kong's public housing sector generally rely on cross-sectional or time series evidence. Wong and Liu (1988) provide evidence on misallocation using data on rent and income in the Population Census. Lui and Suen (2011) study spatial misallocation using mobility patterns, while Cheung et al. (2021) study turnover rates. Yeung (2001) provides descriptive survey evidence and simulations to study how TPS affected Hong Kong's property prices. Ho and Wong (2006) provide time-series evidence on the effects of TPS on private-sector housing prices, but their estimates are potentially confounded by contemporaneous events such as the Asian Financial Crisis.

The paper proceeds as follows. Section 2 describes institutional background. Section 3 provides a theoretical framework. Section 4 presents the estimated effects of TPS. Section 5 discusses economic mechanisms and implications. Section 6 concludes.

2 Institutional Background

2.1 Public Rental Housing in Hong Kong

The purpose of Hong Kong's PRH program is to provide subsidized units for qualifying low-income families. Applicants are funneled through a waiting-list system, which processes applications mainly on a first-come-first-served basis. To receive offers, applicants must satisfy income and asset requirements, which depend on household size.[3] Individual units are then offered to applicants by random computer batching according to each applicant's household size, unit allocation standards, and choice of district. Applicants receive up to three housing offers, which are given out one at a time. If all three offers are rejected, then the applicant must wait one year before reapplying. In 1998, the year before the launch of TPS, 2.3 million Hong Kong residents lived in PRH units, roughly 38 percent of the total population.[4]

The average rent of a PRH unit in 2004 was $1,563 (HKD), about half of a similar private-sector unit.[5] Tenants who lived in PRH units for 10 years or more must declare the income and assets of all household members biennially. Households who reported total monthly incomes in excess of household-size-contingent income limits were required to pay either 1.5 times rent or double rent, and households who additionally had large net asset holdings either paid market rent or were asked to move out.[6] To encourage truthful reporting, income and asset declarations were randomly chosen for in-depth verification. Households with all members aged 60 or above were exempted from income checking.[7]

[3] Appendix Table A1 shows the PRH income limits.

[4] See Housing Department (2021) and Legislative Council Secretariat (2020). As of March 2019, public rental housing units accounted for about 29 percent of the stock of permanent housing and housed about 31 percent of total households in Hong Kong (Census and Statistics Department 2020; Transport and Bureau 2019).

[5] Online Appendix Figure A1 plots rent trends for public housing units and similar units in the private sector.

[6] Appendix Figure A2 shows the PRH rent schedule.

[7] The Housing Subsidy Policy (HSP) and the Policy on Safeguarding Rational Allocation of Public Housing Resources (PSRA) were implemented in 1987 and 1996 respectively and are collectively referred to as "Well-off Tenants Policies". Under the PSRA, household income and net asset value are adopted as the two criteria for determining PRH households' eligibility to continue to receive subsidised public housing. Under section 26(1) of the Housing Ordinance, any person who knowingly makes any false statement are liable on conviction to a maximum fine of $50,000 and to imprisonment for six months. Between 2003 and 2006, roughly 6 percent of households were found to have under-reported their incomes, of which 18 percent were prosecuted. See Audit Commission (2007) for more details.

The government allowed only the tenant and family members listed on the tenancy agreement to occupy a PRH flat. The tenant must notify the government immediately of any household changes caused by birth and death. Upon death of the tenant, the flat may be transferred to the spouse or to an authorized member residing in the flat who satisfies the income limit. The government also reallocates units if the size of a household significantly falls due to move-out, death, marriage, or emigration of household members. However, these cases were rare. Between 2016 and 2020, the government resolved an average of about 2,200 under-occupation cases each year, roughly 0.3 percent of the total number of PRH households.[8]

2.2 History of Tenants Purchase Scheme

In 1997, the Hong Kong Housing Authority announced the Tenants Purchase Scheme (TPS), which allowed PRH tenants to buy the units they lived in at a discounted price. The policy announcement was unexpected and its stated goal was to boost Hong Kong's homeownership rate to 70 percent within ten years' time. Between 1998 and 2006, units in 39 PRH estates, totaling 183,700 units and comprising roughly 27 percent of the total stock of PRH units, were made available for sale. The sale proceed in five rounds, starting every two years. In each of the five rounds, around 26,000 to 28,000 PRH units in six selected estates were offered for sale. In the last round, which comprised phase 6A and phase 6B, around 49,000 PRH units in nine estates were offered for sale (Legislative Council Secretariat 2020). In August 2005, the Housing Authority announced that no additional estates will be put up for sale under the TPS program after 2006. Section 4 leverages the staggered and incomplete roll-out of TPS across housing estates to identify the impact of the program.

[8] To address under-occupation (UO), tenants are required to declare biennially their occupancy position. These declarations are verified through random unit visits. If the number of household members in a PRH unit is below the minimum number set by the HA for the unit, the household is asked to move to a suitable unit. Under-occupation is a significant problem. As of March 2021, there were 79,380 UO households, of which 5,320 were considered prioritized UO cases. See Audit Commission (2013) and GovHK (2021).

2.3 Restrictions on Resale and Leasing of TPS Units

TPS granted a peculiar form of occupancy right to purchasing households. TPS unit owners were no longer subject to the regular income tests and under-occupancy unit allocation rules of PRH tenants. Therefore, TPS owners and their registered family members can occupy the purchased unit indefinitely. TPS owners were also permitted to transfer ownership to registered family members in special circumstances, such as old age or illness. At the same time, family members who once resided in TPS units but become unregistered were eligible to apply for PRH units. As such, it became possible for a family member with high earnings potential to gain ownership of the unit without satisfying income tests, while a family member with low earnings potential applied for another subsidized unit.

Strong incentives were put in place to encourage rapid sale. Almost all sitting tenants in the selected estates were offered the opportunity to purchase.[9] Tenants who do not wish to purchase can continue to rent and occupy their units as before. The purchase price was set at replacement cost, but given a further discount of 60% on purchase within the first year, which is as low as 12% of market value.[10] Following the sale, the unit owner became responsible for maintenance and repairs, building management fees, as well as property taxes.

However, TPS owners were restricted from resale and letting. First, they cannot lease or resale on the open market until a premium equivalent to the current value of the original discount is paid to the government.[11] Second, TPS owners were permitted to sell their unit without payment of a premium only to public housing renters and a small number of eligible purchasers

[9] The exceptions were those living in the following units: 1) Housing for Senior Citizens and Small Household Block; 2) units used for social welfare purposes; and 3) units with common entrance and communal facilities such as bathroom, kitchen and entrance.

[10] New tenants who purchase TPS units enjoy a full credit if they buy within the first year and a halved credit in the second year. After the second year, no credit will be given. Purchasers will need to pay, apart from the price of the unit, the stamp duty, registration fees and legal costs. See Housing Authority (2014) for more details.

[11] In the first two years after the sale, a TPS unit owner can only sell the unit back to HA at the list price. Within the third to fifth years from the date of first assignment, TPS unit owners can sell back their units to Housing Authority at assessed market value less the original purchase discount. If HA declines to buy back the units, however, TPS unit owners can sell, let or assign their units in the open market. In addition, the Housing Authority may give consent to a request for change of ownership under special circumstances, such as divorce or separation, emigration or long-term working abroad, death, old age, bankruptcy, or terminal illness of owner. TPS owners letting units in breach of the Housing Ordinance are liable on conviction to a maximum fine of $500,000 and to imprisonment for one year. See Housing Authority (2014).

in the Home Ownership Scheme (HOS) Secondary Market. Both types of transactions were rare, as shown below in Section 5.2.

3 Theoretical Framework

In this section, two simple models are developed to illustrate the impacts that TPS may have had on the behavior of public housing residents. The first model characterizes how TPS affected labor supply and household-level sorting. The second model considers how TPS affected co-residence decisions within family networks.

3.1 Model with Atomistic Households

Consider a set of public housing units and a set of atomistic households with wage distribution $w \sim F$. Households have utility $u(c,h,l) = c^{\alpha} h^{\beta} l^{\gamma}$ over consumption c, housing services h, and leisure l, with $\alpha + \beta + \gamma = 1$. The timing is as follows. First, households are randomly assigned to identical public housing units, a random subset of which are TPS-eligible. Second, households choose between residing in public or private housing. Third, TPS-eligible public renters choose between renting and purchasing their units. We assume there is an excess of households willing to live in public housing units, so they are all filled.

If renting in the private sector, the household's utility is:

$$u_{Pri}(w) = \max_{c,h,l} u(c,h,l) \quad \text{s.t.} \quad c + rh \leq w(T-l),$$

where r denotes the rent per housing service in the private sector, and w denotes wage.

For residents of public rental housing, the quantity of housing services is fixed at $\bar{h}$. The utility of a household in public rental housing is:

$$u_{PRH}(w) = \max_{c,l} u(c,\bar{h},l) \quad \text{s.t.} \quad c + R(w(T-l)) \leq w(T-l).$$

where public housing rent is given by $R(I)$, an income-contingent rent schedule with two notches:

$$R(I) = \begin{cases} \overline{R} & \text{if } I < 2\bar{I} \\ 1.5\overline{R} & \text{if } I \in [2\bar{I}, 3\bar{I}] \\ 2\overline{R} & \text{if } I > 3\bar{I}. \end{cases}$$

We assume that $2\overline{R} + < r\overline{h}$, so public housing rent is always lower than the private sector.

If they purchase the public housing unit through TPS, then the household's utility is given by

$$u_{TPS}(w) = \max_{c,l} u(c, \overline{h}, l) \quad \text{s.t.} \quad c + \overline{m} \leq w(T - l),$$

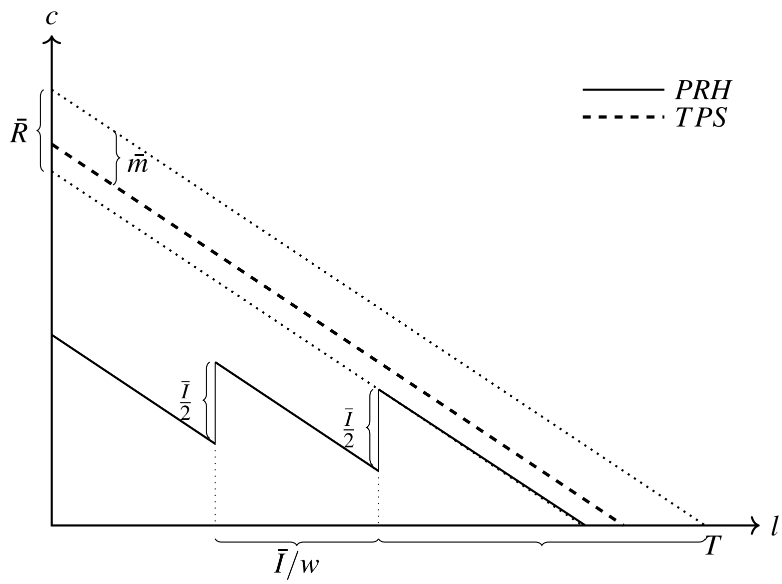
where $\overline{m}$ denotes the cost of residence for TPS owners. As explained in Section 2, TPS allowed sitting PRH tenants to purchase permanent occupancy rights to their units at heavily subsidized prices but did not grant leasing or resale rights. Therefore, compared to PRH tenants, the budget set of a TPS household is altered in two ways: (1) housing costs do not depend on income (i.e. rent notches are removed); and (2) non-income-contingent housing costs may also be different (i.e., $\overline{R} \neq \overline{m}$).

Figure 1 compares the budget sets of PRH and TPS households. Note that the budget set of PRH households closely resembles that of households who choose between leisure and consumption in the presence of income tax notches. TPS removes the rent notches, yielding a linear budget line. Analytic derivations of each household's optimal consumption, leisure, housing, income and indirect utility are presented in Online Appendix B. For simplicity, I discuss here only the case where $\overline{m} \leq \overline{R}$, where all eligible households purchase TPS units.[12]

First, consider the case where $\bar{m} = \bar{R}$. Figure 2 simulates the housing consumption, labor supply, income, and utility of households in this case, under different housing arrangements, as a function of their wage. In this case, the rent for low-income households is the same for PRH and TPS households. The difference between PRH and TPS households is that rents discontinuously increase at the income cutoffs $2\bar{I}$ and $3\bar{I}$ for PRH residents, but not for TPS

[12] If $\overline{m} > \overline{R}$, then some households may prefer not to purchase TPS units, and only a subset of households in TPS-eligible estates are affected by TPS.

Figure 1: Budget set of PRH and TPS households



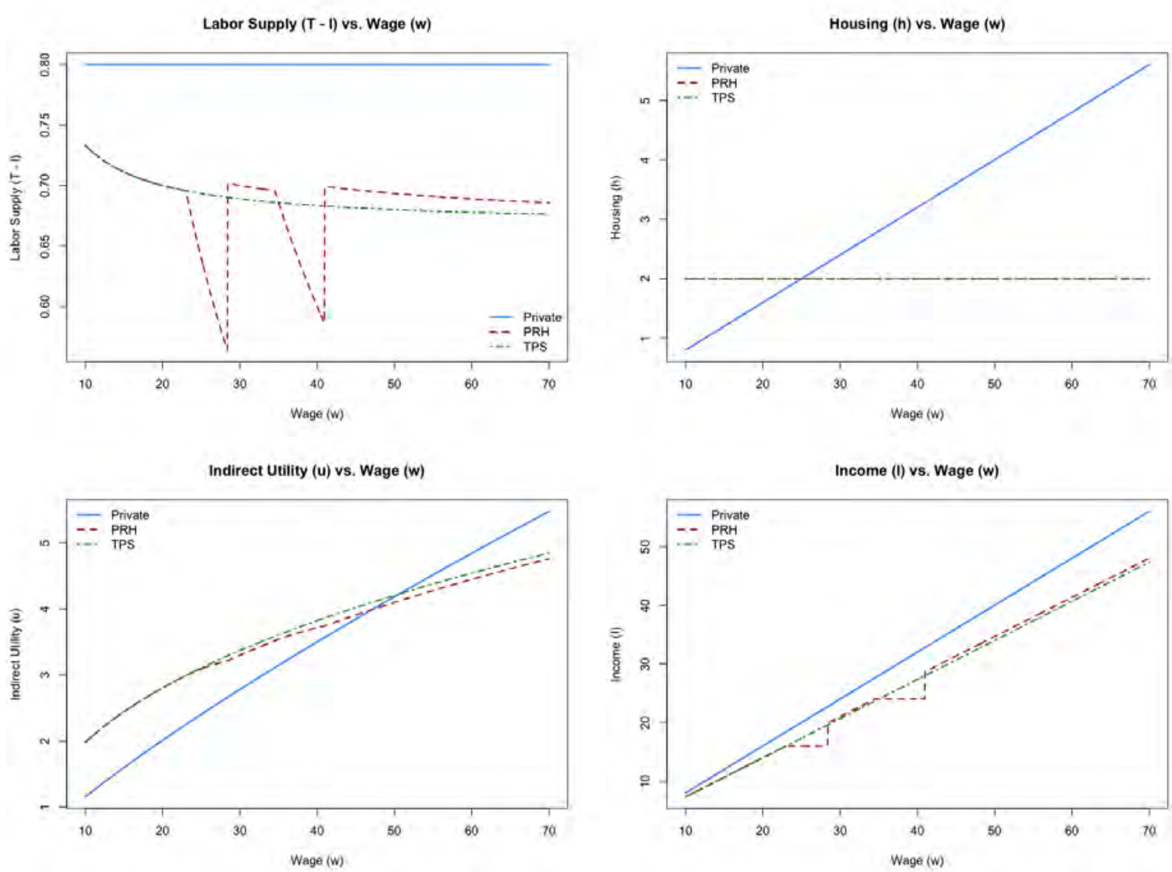
residents.

As shown in the figure, the PRH rent "notches" have two impacts. First, the notches disincentize households in some wage ranges from working. Specifically, around the notch, a small increase in income will lead to a discontinuous jump in rent, so optimizing PRH households reduce labor supply in such ranges to avoid the increase in rent. Second, the notches encourage a subset of higher-wage households to sort into private housing. To see this, note that PRH rent notches reduce the utility of higher-wage PRH households, who must pay extra rent to receive public housing. Higher-wage PRH households are thus more likely to prefer private-sector living than TPS residents with similar wages.

The case with $\overline{m} < \overline{R}$ is similar. As above, TPS removes rent notches. In addition, it induces a reduction in housing costs for all households (since $\overline{m} < \overline{R}$). As shown in Figure 1, this reduction in housing costs is equivalent to an outward shift of the budget set for households residing in public housing units. The result is both reduced labor supply and a weakened preference for high-wage households to live in private-sector units.

The difference in average incomes between TPS-eligible and non-TPS-eligible public hous-

Figure 2: Labor supply, housing, income, and utility by household wage, by housing types



Simulation parameters: $T = 1; \bar{I} = 8; \bar{R} = 2; \alpha = 0.4; \beta = 0.4; \gamma = 0.2; \bar{h} = 2; r = 5; \bar{m} = 2.$

ing residence is given by:

$$\Delta = \mathbb{E}[I_{TPS}(w_i) \mid w_i < w_{TPS}] - \mathbb{E}[I_{PRH}(w_i) \mid w_i < w_{PRH}] \tag{1}$$

where $I_{TPS}(w_i)$ denotes the chosen income of household i if they purchased a TPS unit, and $I_{PRH}(w_i)$ if they rented a PRH unit. As Equation (1) makes clear, the average income of TPS residents is different from that of PRH residents for two reasons: (i) TPS residents have higher average wages, since $w_{TPS} > w_{PRH}$, and (ii) conditional on wage, TPS resident may have a different level of labor supply, due to the removal of the rent notch.

3.2 Model with Co-residence Choices

This subsection explores how TPS affects co-residence patterns. In both PRH units and TPS units with unpaid premiums, government rules allow family members (such as parent, child, and grandchild) to co-reside, but otherwise restrict leasing and resale. Therefore, by removing regular income tests, TPS may shift households towards being composed of fewer extended family members and those with higher earning power.

A Shapley–Shubik–Becker matching framework is used to model co-residence choices. For simplicity, incomes are now assumed to be fixed. We consider two people in an extended family, labeled L and H, where H is the higher earner, with income $I_H > I_L$. L and H jointly have access to a PRH unit. The two choose whether to live together in the PRH unit or to live separately. If they live separately, the person who does not live in the assigned PRH unit may live in either the private sector or another PRH unit.

If an individual lives alone in a PRH unit (call her i), her utility is

$$v_{PRH}(I_i) = \max_c u(c,\overline{h}) \quad \text{s.t.} \quad c + R_1(I_i) \leq I_i,$$

where $u(c,h) = c^a h^{1-a}$ and R_n is the income-contingent rent schedule for an n-person household.[13] If she lives in a private-sector unit alone, then her utility is

$$v_{Pri}(I_i) = \max_c u(c,h) \quad \text{s.t.} \quad c + rh \leq I_i.$$

If instead L and H co-reside in the PRH unit, then their joint utility is

$$\begin{aligned} V_{CoPRH} = &\max_{c_L,c_H} \frac{1}{\delta}[u(c_L,\overline{h}) + u(c_H,\overline{h})] \\ &\text{s.t. } c_L + c_H + R_2(I_L + I_H) \leq I_L + I_H, \end{aligned}$$

where $\delta > 1$ is a congestion cost associated with shared residence.

Co-residence decisions are determined by maximizing joint utility. For simplicity, it is

[13]Here we let $R_n(I) = \overline{R}\left[1 + \frac{1}{2}1\{I > 2\overline{I}_n\} + \frac{1}{2}1\{I > 3\overline{I}_n\}\right]$.

assumed that $v_{PRH}(I_L) > v_{Pri}(I_L)$, so L prefers a PRH unit to a private unit when living alone. There are three undominated possibilities for PRH families: (i) co-residing in the PRH unit; (ii) living separately, in respective PRH units; and (iii) living separately, with L in the PRH unit and H in the private sector. The respective joint utilities in each case are (i) V_{CoPRH}, (ii) $V_{PRH,PRH} = v_{PRH}(I_L) + v_{PRH}(I_H)$, and (iii) $V_{PRH,Pri} = v_{PRH}(I_L) + v_{Pri}(I_H)$.

Now consider a household that has purchased a TPS unit and hence is no longer subject to income tests. If residing in the TPS unit alone, the household member now has utility

$$v_{TPS}(I_i) = \max_c u(c, \overline{h}) \text{ s.t. } c + \overline{m} \leq I_i.$$

If they co-reside, their joint utility is

$$\begin{aligned} V_{CoTPS} = &\max_{c_L, c_H} \frac{1}{\delta}[u(c_L, \overline{h}) + u(c_H, \overline{h})] \\ &\text{s.t. } c_L + c_H + \overline{m} \leq I_L + I_H, \end{aligned}$$

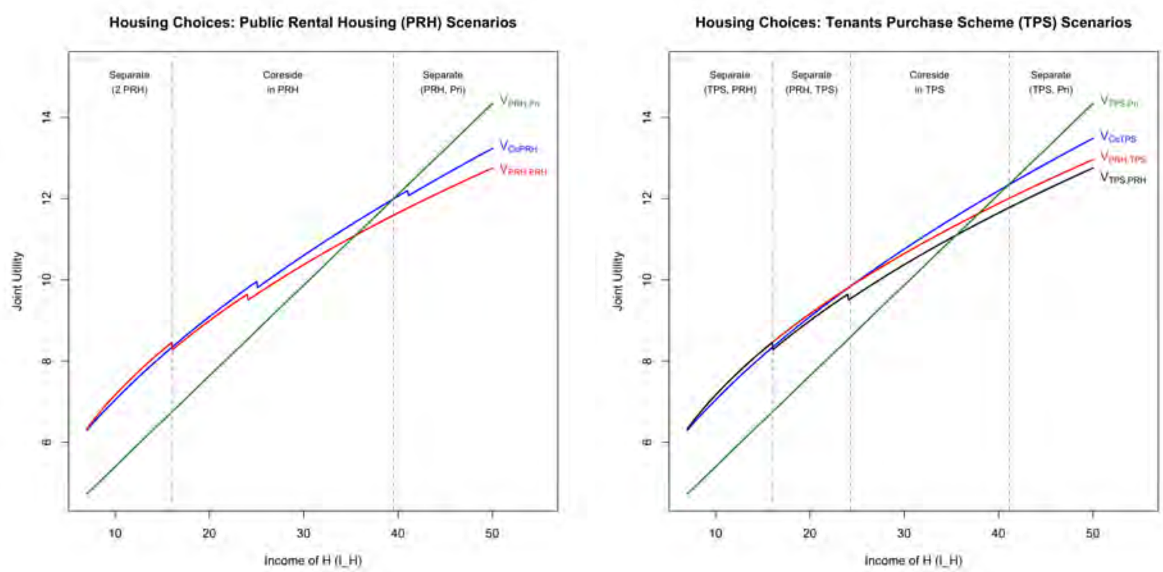
For TPS families, there are four undominated possibilities: (i) co-residing in the TPS unit; (ii) living separately, with L in the TPS unit and H in the private sector; (iii) living separately, with L in the TPS unit and H in a PRH unit; and (iv) living separately, with L in a PRH unit and H in the TPS unit. The respective joint utilities in each case are (i) V_{CoTPS}, (ii) $V_{TPS,PRH} = v_{TPS}(I_L) + v_{PRH}(I_H)$, (iii) $V_{TPS,Pri} = v_{TPS}(I_L) + v_{Pri}(I_H)$, and (iv) $V_{PRH,TPS} = v_{PRH}(I_L) + v_{TPS}(I_H)$. Analytic derivations are presented in Online Appendix C.

Figure 3 simulates the joint utilities, as a function of I_H, holding I_L constant. It is shown that in certain income ranges, it is optimal for family members to co-reside in the public housing unit under PRH, but not under TPS. As such, TPS resident households are likely to have smaller household sizes.

3.3 Empirical Implications

The models above show that TPS has three potential impacts: (i) it alters the labor supply of those living in public housing, (ii) it discourages high-wage individuals moving to the private-

Figure 3: Joint utility by income of higher earner, for PRH and TPS households



Simulation parameters: $I_L = 7; \bar{I}_1 = 8; \bar{I}_2 = 16; \bar{R} = 2; a = 0.5; \bar{h} = 2; r = 5; \bar{m} = 2; \delta = 1.1$.

sector, and (iii) it alters co-residence decisions within extended family networks, potentially by encouraging high-income members to stay and low-income members to obtain another subsidized unit. Section 4 estimates the impact of TPS on the distribution of household incomes, measured as multiples of the household-size-contingent PRH income limit. Section 5 assesses the relevance of each above three channel through examining auxiliary evidence and the impacts of TPS on additional outcomes.

4 Effects of TPS on Estate-level Outcomes

This section estimates the effects of TPS using its staggered and incomplete rollout across housing estates. The estimates reveal that TPS dramatically increased average household income.

4.1 Empirical Strategy

To measure the effects of TPS on estate outcomes, restricted-access data from the Hong Kong Population Census and By-census are used—specifically, the 20% random samples in 2001, 2011 and the 10% random samples in 1996, 2006, 2016. These data provide information about each respondent's age, sex, household composition, employment, and earnings, as well as an indicator for whether the respondent moved in the last five years.[14] These data also include identifiers for 136 public rental housing estates, including all 39 estates where residents became eligible to partake in TPS. The identifiers are used to construct a panel of treated and control housing estates for measuring the causal impact of TPS.

Although documentation regarding how the treated estates were chosen by the authorities is not publicly available, conversations with relevant officials suggest that more recently constructed estates were chosen in hopes of encouraging rapid sale. To form the control group, estates with buildings constructed before 1980 are excluded, so that treated and control estates had more similar building features and resident populations. Estates with any buildings constructed after 1996 are also excluded, so that our estimates are not contaminated by influxes of new residents upon the completion of new construction.[15] This leaves 39 treated estates and 43 control estates.

To identify the effects of TPS on estate-level outcomes, I leverage the staggered and incomplete roll-out of the program across estates in a dynamic difference-in-differences design. Specifically, I estimate the following equation:

$$y_{et} = \sum_{\tau \in \mathscr{T}} \beta_\tau \left(T_e \times 1_{t=t_e^*+\tau} \right) + \delta_e + \delta_t + \gamma_t X_e + \varepsilon_{et},$$

where e indexes estates, $t \in \{1996, 2001, 2006, 2011, 2016\}$ is the Census year, y_{et} is an estate-level outcome variable, T_e indicates whether estate e was ever treated, t_e^* is the first Census year following treatment for estate e, $\tau \in \mathscr{T} \equiv \{-10, 0, 5, 10, 15\}$ indexes the year relative to t_e^*, δ_e and δ_t denote estate and year fixed effects, and X_e denotes estate characteristic. This equation includes year fixed effects and thus controls for confounding city-wide changes in the housing market that contaminates previous estimates of the effects of the TPS program (e.g.

[14]Real income is deflated using 1996 dollars.

[15]Online Appendix Table A2 and A3 displays the sample restrictions and lists the chosen estates. Building construction years are collated from four sources: (1) data.gov.hk; (2) Wikipedia; (3) website of the Housing Society; and (4) website of the Housing Authority.

Ho and Wong 2006). It also includes time-varying estate-characteristic controls, to account for potential imbalances between treatment and control estates. The β_τ coefficients identify the causal effect of TPS under the assumption that the outcomes of treated estates would have evolved in parallel to those of control estates in the absence of treatment, having accounted for observable imbalances.

The lack of a longer panel limits my ability to check for pre-treatment differential trends that may arise from unobserved imbalances. Below it is shown that pre-event differential trends are not detected in the observed pre-event years. However, estate-level outcomes are observed for at most two pre-treatment Census years, and with two years observed for only 18 out of the 39 treated estates.

The first important threat to causal identification is the possibility that initial differences in demographic composition may cause the regression estimates to pick up "natural" life-cycle changes in income and household composition. To assess this possibility, the pre-treatment demographic and spatial composition of the treatment and control estates are shown in the first three columns in Table 1. Each estate houses roughly 4,500 households, or a population of roughly 18,000. Their average household incomes and average rents are highly similar. However, treated estates have larger average household sizes and somewhat different demographic composition, with treated estates featuring a much larger share of younger birth cohorts.

To make treated and control estates more comparable, the regression sample is reweighted to be similar in terms of initial demographic composition using inverse propensity score (IPS) weights. Propensity scores are formed from a logistic regression of treatment status on estate characteristics, including construction year, average household size, and population shares of birth cohorts. The latter three columns in Table 1 shows that the differences in birth cohort composition are much smaller and largely insignificant after reweighting. In the main specification, I report estimates using a combination of weighting and regression adjustment (i.e., adding region-specific and construction-period-specific trends controls for inverse propensity scores), a method known to be "doubly robust" to misspecification (Glynn and Quinn 2010). In addition, estate construction years are binned into five-year intervals, and bin-specific trend controls are added, to account for differences in population and housing characteristics associated with

Table 1: Estate HH composition, treated vs control estates, 1996

	Unweighted sample			Weighted sample		
	Treated estates	Control estates	t-stat	Treated estates	Control estates	t-stat
Year built	1989	1986	2.51	1989	1987	2.02
	(1.7)	(5.3)		(2.3)	(5.1)	
Average HH size	4.0	3.7	3.98	4.0	3.8	3.26
	(0.3)	(0.4)		(0.2)	(0.4)	
Average HH income	16221	16323	-0.18	17095	16676	0.99
	(2782)	(2307)		(2473)	(2428)	
Cohort						
1920s	0.05	0.06	-1.82	0.05	0.06	-1.47
1930s	0.07	0.09	-2.53	0.08	0.08	-0.97
1940s	0.08	0.10	-3.63	0.09	0.10	-0.79
1950s	0.18	0.17	1.29	0.17	0.17	-0.41
1960s	0.17	0.15	1.79	0.16	0.16	0.66
1970s	0.14	0.18	-3.83	0.17	0.17	-0.59
1980s	0.21	0.16	4.08	0.18	0.16	2.09
1990s	0.08	0.07	1.18	0.07	0.07	0.05
Region						
Urban	0.33	0.44	-1.00	0.40	0.48	-0.97
Extended urban	0.33	0.28	0.53	0.36	0.24	1.52
New territories	0.33	0.28	0.53	0.25	0.28	-0.48
Number of estates	39	43		39	43	

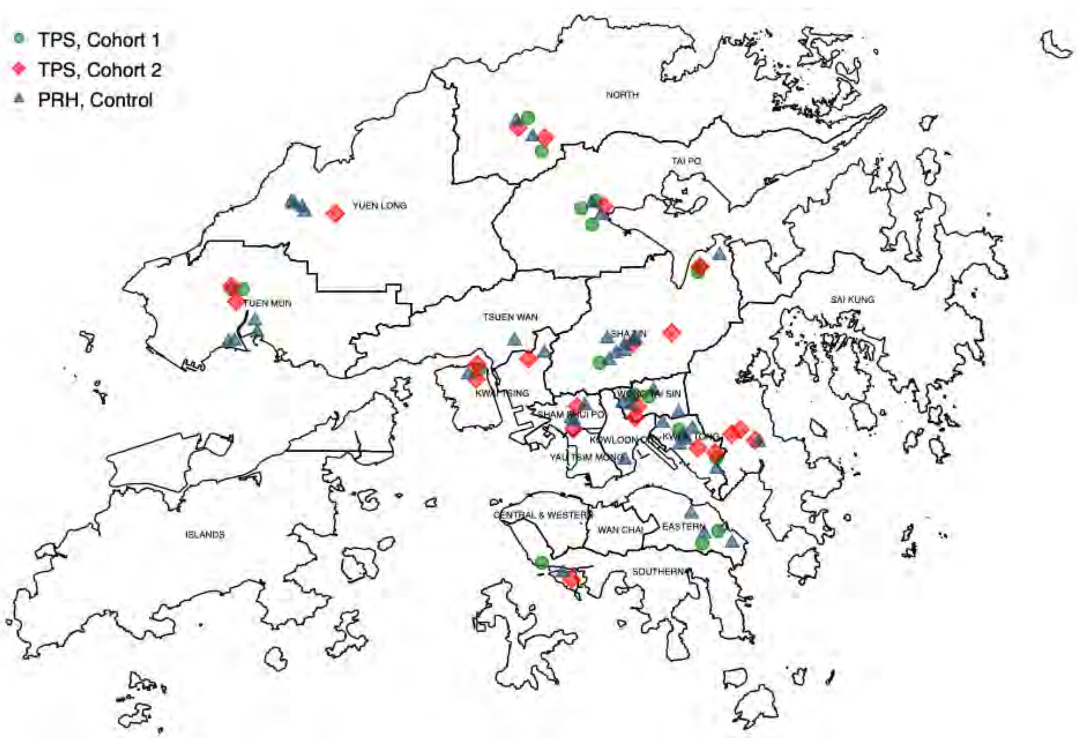
Notes: Table shows mean estate characteristics in 1996, of unweighted sample and sample with inverse propensity score weighting, respectively for TPS and non-TPS estates.

construction year. Estimates without the weights and additional controls are also reported and shown to be similar.

Another potential threat is spatial clustering in treatment assignment. The last few rows in Table 1 suggests that there does not appear to be significant spatial clustering. Figure 4 displays a map of the estates, revealing that the treated and control estates are evenly dispersed across Hong Kong. In our main regression specification, time-varying controls for urban, extended urban, and rural regions are added to account for potential spatial differences across estates.

Since TPS introduction was staggered across estates, my main specification uses the esti-

Figure 4: Map of treated and control estates



Notes: Figures plots each treated and control estate included in the analysis sample.

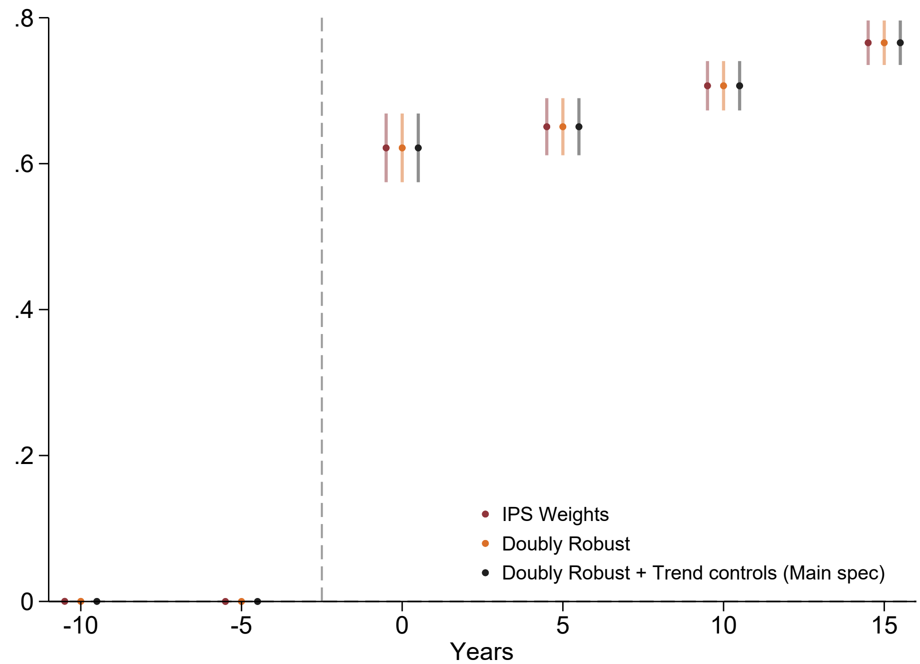
mator proposed by Borusyak, Jaravel and Spiess (2024).[16] Alternative estimators designed for staggered difference-in-differences designs, proposed by Callaway and Sant'Anna (2020) and Sun and Abraham (2020), are used in robustness checks. Standard errors are clustered at the estate level.

4.2 TPS Take-up

Figure 5 shows that the share of households residing in sold TPS units immediately rose by 62 percent once residents became eligible to purchase TPS units in Year 0. This share eventually reached 77 percent higher than control in Year 15. The top columns in Appendix Table A7 confirms the overall trend in take-up of households in TPS estates. It shows that a large majority

[16]This specification ensures that estimates are not contaminated by treatment effects from other periods when treatment is staggered (Callaway and Sant'Anna 2020; de Chaisemartin and D'Haultfœuille 2020).

Figure 5: Effects of TPS on share of HHs with purchased TPS unit



Notes: Figure plots the effect of TPS on the share of households with purchased TPS unit. The black series plots coefficients estimated using the interaction-weighted specification of Borusyak, Jaravel and Spiess (2024), augmented with a doubly robust procedure, region-specific and construction-period-specific trends control. The maroon plots coefficients, estimated with only inverse propensity score weights, while yellow series are estimated only with a doubly robust procedure. Inverse propensity score weights are constructed by first estimating the probability of treatment at the estate level using a logistic model with the following covariate: year in which the estate was built, average household size, and the shares of residents in each birth cohort in the pre-treatment Census year. Sample is all estates where all buildings were built after 1979 and before 1996. Year 0 denotes the first observed Census year following treatment. Confidence intervals at the 95% significance level (clustered at the estate level) are shown in bars. Online Appendix Table A5 displays coefficients and pre-treatment means.

of units in TPS estates were sold immediately after the launch of TPS. By 2006, the share of households residing in sold TPS units had risen to 57.4 percent from zero in 1996. By 2016, the share further increased to 71.9 percent.

Which households took up TPS? A government study in 2001 reported that "the sale results of TPS flats were better among households who were paying additional rent, of larger size and with non-elderly members" (Housing Authority 2001). Online Appendix Table A4 confirms that larger and higher-income households were more likely to live in sold TPS units. Households whose members are all over 60 years old and therefore not subject to means testing requirements were less likely to live in sold TPS units.

Why were higher-income households more likely to take up TPS? One possibility is that high-income household were better able to obtain loans. However, this channel is somewhat weakened by the fact that the government agreed with several banks to provide mortgages of up to 100% of the balance of the purchase price of the unit for up to 25 years. Instead, Yeung (2001) presents survey evidence that fear of paying extra rent was an important motivator for TPS purchases. Anecdotal evidence also provides support for the proposition that avoidance of regular income tests was a major motivation for households to purchase TPS units. For example, Legislative Council member Wong Kwok-kin stated on record that "[m]any well-off tenants want to buy their own flats through the TPS so as to avoid the trouble of paying double rent or undergoing random checking"(GovHK 2012).

4.3 Effects of TPS on Estate-level Income Distribution

Having characterized the drivers of TPS take-up, this subsection asks: How did TPS affect the household income distribution in treated estates?

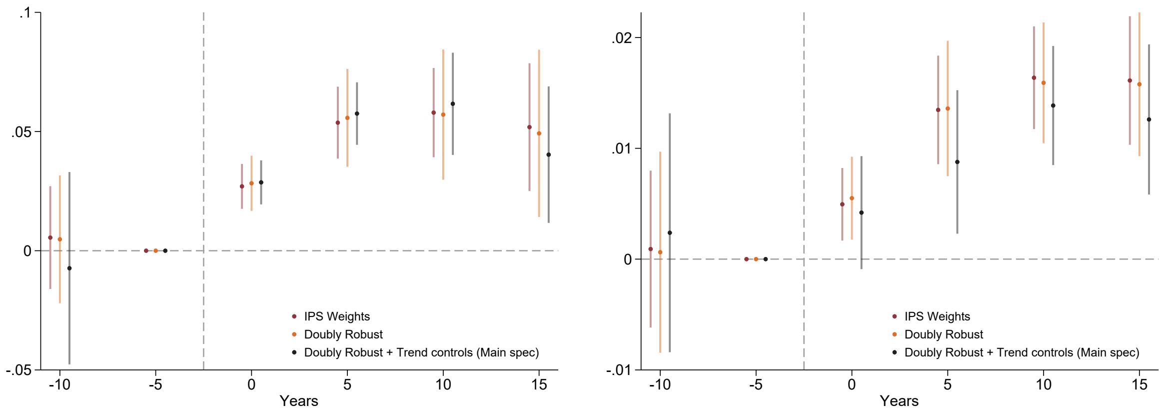
Figure 6 shows the main result. Panel (a) shows that the share of households above the 2X PRH income limit rose sharply. By Year 0, the share of households above 2X PRH income limit increased by 2.7 percentage points, or 25 percent relative the 1996 mean in treated estates of 11.4 percent. This divergence further widened thereafter. By Year 15, the share of households above the 2X PRH income limit limit was 5.0 percentage points (or 44 percent) higher than control.

Panel (b) shows a similar pattern exists for the share of households above the 3X PRH income limit. By Year 0, the share of households above 3X increased by 0.5 percentage points (or 19 percent) in treated estates. By Year 15, the share of households above 3X increased by 1.6 percentage points (or 59 percent).

Online Appendix Figure A4 shows that the above estimates are highly robust to the use of alternative difference-in-differences estimators. Online Appendix Figure A5 shows that the results are also robust to using no weights and using alternative IPS weights based only on estate

Figure 6: Effect of TPS on HH income distribution relative to 2X and 3X PRH income limit

(a) Share of HHs above 2X PRH income limit (b) Share of HHs above 3X PRH income limit



Notes: Figure plots the effect of TPS on the share of households within a given household income bin relative to the 2X PRH income limit and 3X PRH income limit. Other notes follow Figure 5.

construction year and share of households within household size bins. This insensitivity to alternative specifications suggests that the increases are unlikely to be attributable to underlying trajectories rather than TPS.

Figure 7 plots the effect of TPS on the share of households within household income bins.[17] The figure reveals that the share of households with incomes much lower than the 2X PRH income limit dramatically fell in treated estates, the share of households just below did not change, while the share of households with incomes just above the PRH limit sharply increased.[18]

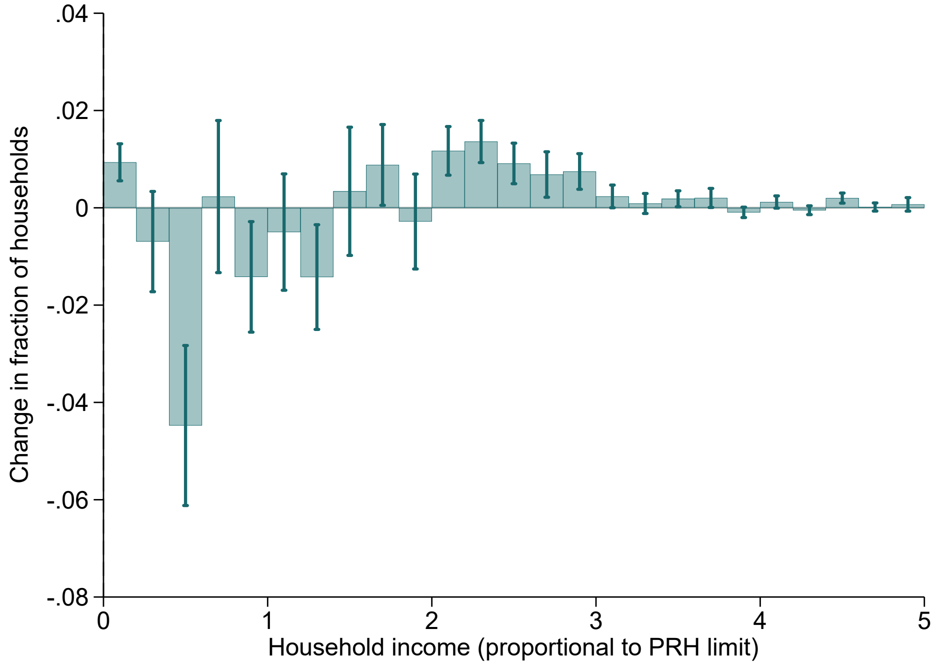
5 Mechanisms and Implications

This section documents the impact of TPS on estate-level auxiliary outcomes, to assess why TPS increased the share of households with incomes above PRH income limits in treated estates. First, the impacts of TPS on average household incomes and household sizes are reported.

[17] This exercise relates to a growing literature on bunching at tax kinks, tax notches, and wage floors (Saez 2010; Kleven and Waseem 2013; Kleven 2016; Cengiz et al. 2019; Blomquist et al. 2021).

[18] Online Appendix Figure A3 plots the raw income distributions of the treated and control estates, by household size, over time, showing a similar right-ward shift in treated estates.

Figure 7: Effect of TPS on HH income distribution



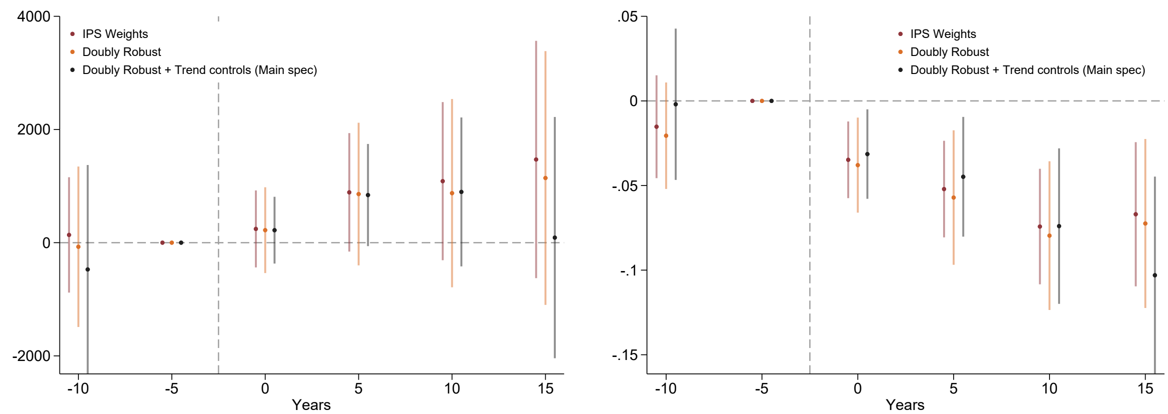
Notes: Figure plots the effect of TPS on the share of households within a given household income bin in the second Census following treatment relative to that of the last Census year before treatment, estimated using the interaction-weighted estimator in Borusyak, Jaravel and Spiess (2024), augmented with a doubly robust procedure, region-specific and construction-period-specific trends control. Income is normalized to be a proportion of the PRH income limit for the relevant household size. Confidence intervals at the 95% significance level (clustered at the estate level) are shown.

Second, the impacts of TPS on household sorting and composition are reported. Third, the effects of TPS on average income, employment, schooling, and commute times within demographic groups are reported. Fourth, the effects of TPS on average housing expenditures are shown.

The conceptual framework in Section 3 suggested that TPS may increase the share of households above PRH income limits through three channels: (i) household-level resorting, (ii) labor supply changes, and (iii) within-household co-residence changes. Although only estate-level aggregate outcomes are available, the evidence is strongly suggestive that TPS caused households to strategically alter co-residence choices in order to take advantages to loosened income requirements. As shown below, household-level resorting was limited. Changes in labor supply

Figure 8: Effects of TPS on HH incomes and HH size

(a) Average HH income (b) Log average HH size



Notes: Figure plots the effect of TPS on average real household income and log average household size. Household income is deflated by Hong Kong CPI based on 1996. Other notes follow Figure 5.

are also unlikely to fully explain the observed income increases. A back-of-the-envelope calculation suggests that by encouraging less eligible households to reside in treated estates, TPS may have substantially reduced public housing access and exacerbated the subsequent subdivided unit crisis.

5.1 Effects on Household Income and Household Size

The main result in Section 4.3 is that TPS reduced the share of households with incomes above PRH income limits. Is this attributable to increased incomes or reduced household sizes?

Figure 8, Panel (a) shows no statistically significant change in the average household incomes. In Year 0, average household income rose $171 (HKD), or 1 percent higher. This increase widened to $1182 higher (or 7 percent higher) than control by Year 15. These estimates, however, are statistically insignificant.

Figure 8, Panel (b) shows that average household size in treated estates immediately declined. In Year 0, average household size fell by 3.8 percent. This decline widened in the earlier post-treatment years, but it contracted to 7.1 percent lower than control by Year 15. These declines are highly significant.

Online Appendix Figure A6 assesses how these estimates depend on the use of regression weights. Panel (a) shows that without adjusting for initial differences in population composition, a much larger increase in household income is found. Panel (b) shows that the unweighted regression yields a more modest decline in household size, whereas both the main specification shows a larger reduction.

Online Appendix Table A6 Column (6) shows that total population in treated estates declined by 5.9 percent in Year 0. This reduction was persistent and reached 10.3 percent lower than control in Year 15. Since the total population in TPS estates in 1996 was roughly 733,000, the estimates imply that the population in TPS estates fell by roughly 75,000.

5.2 Effects on Household-level Mobility and Household Composition

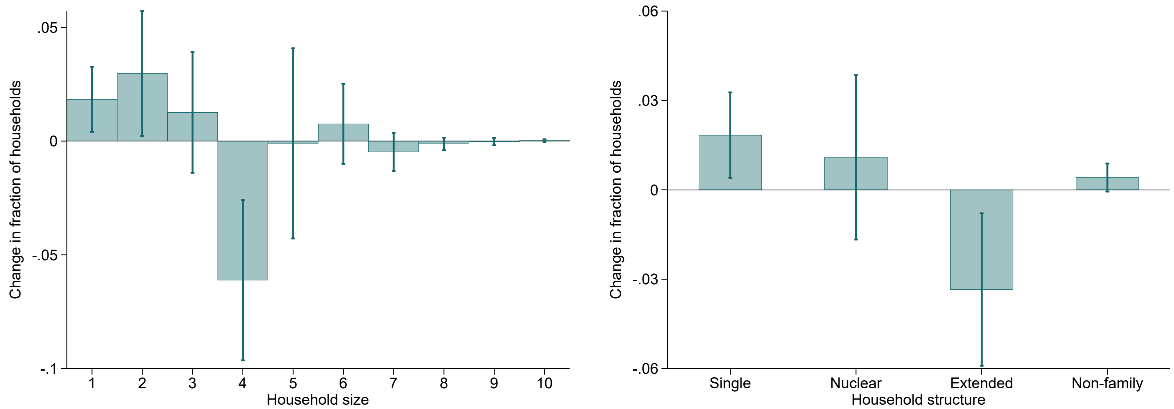
Did TPS cause entire households to move, or alter co-residence choices within households? Evidence suggests that household-level mobility was likely very limited. Although individuals cannot be traced across Census waves, Online Appendix Figure A7 plots the trend in the share of household who reported that they moved in the past five years. Since these estates were constructed before 1996, a large quantity of move-ins is found prior to 1996. In 2001, however, the share of household heads who moved in the past five years was roughly 12 percent. Moreover, there was no detectable change in household-level mobility in the treated estates thereafter, relative to the control estates, as shown in Online Appendix Table A6 Column (4).

Why was residential mobility so low? Online Appendix Table A7 shows that nearly 99 percent of sold TPS units were owner-occupied with their premium unpaid, even until 2016. This implies that only a tiny proportion of sold TPS units were either rented out or resold on the open market, since the premium must be paid before a TPS owner could sell, let, assign, or otherwise alienate the unit on the open market. For example, in the district of Tuen Mun, there were 14,383 sold TPS unit as of September 23, 2021, of which only 200 had premiums paid between 2005 and 2020. In other words, the number of premium payments per year was less than 0.1 percent of the stock of sold TPS units.[19] To see why premium payments were rare, suppose that a unit was purchased at 12 percent of the initial market value, and the household

[19] See: https://www.housingauthority.gov.hk/en/home-ownership/information-for-home-owners/premium-payment-arrangement/premium-statistics/index.html

Figure 9: Effect of TPS on distribution of household types

(a) HH size (b) HH type



Notes: Figure plots the effect of TPS on the share of households with a given household size or type in the second Census following treatment relative to that of the last Census year before treatment. Single households include only one person. Nuclear households include a couple and any of their children. Extended-family households include a nuclear family and additional relatives, e.g. at least one parent of the couple. Other notes follow Figure 7.

now wishes to sell the unit on the open market and simultaneously purchase another unit of equivalent value on the open market. The premium requirement is then equivalent to an 88 percent transaction levy. This requirement strongly discouraged premium payment.

Even without payment of a premium, TPS owners can sell their unit to public housing renters and a small number of eligible purchasers in the Home Ownership Scheme Secondary Market (HOSSM). However, the number of transactions in HOSSM was also small. For TPS units in the district of Tuen Mun, there were only 702 such transactions between the beginning of 2002 and October 2021. The number of transactions in the HOSSM per year was therefore less than 0.3 percent of the stock of sold TPS units.[20] One reason for the low number of transactions is that most eligible purchasers can rent or wait to buy from the government at subsidized rates and therefore had low willingness to pay. Meanwhile, TPS owners were generally unwilling to sell at discounted prices, since they are ineligible to purchase in the secondary market and would not be able to obtain a unit of equivalent value in the open market.

If household-level mobility was limited, how did TPS affect household composition? Figure

[20]See: https://www.housingauthority.gov.hk/en/home-ownership/hos-secondary-market/transaction-records/index.html

9 Panel (a) shows that in Year 5, the shares of households with one or two members rose, while the shares of households with four or five members fell, whereas the shares of other household sizes remained largely unchanged. Panel (b) shows that the share of extended-family households fell by 3.5 percentage points, while the share of single family households rose by 2 percentage points, respectively. Online Appendix Figure A8 shows that TPS reduced populations born in 1930s, 1960s and 1990s in the treated estates. Online Appendix Figure A9 Panel (a) shows that the share of younger women who are single, married, and divorce are unchanged. Panel (b) shows that the average number of children of younger adult residents also unchanged.[21]

5.3 Impacts on Labor Market Characteristics by Groups

Having shown that household sizes fell even as household incomes stayed constant, it is now asked: Why did TPS increase average labor market characteristics in treated estates?

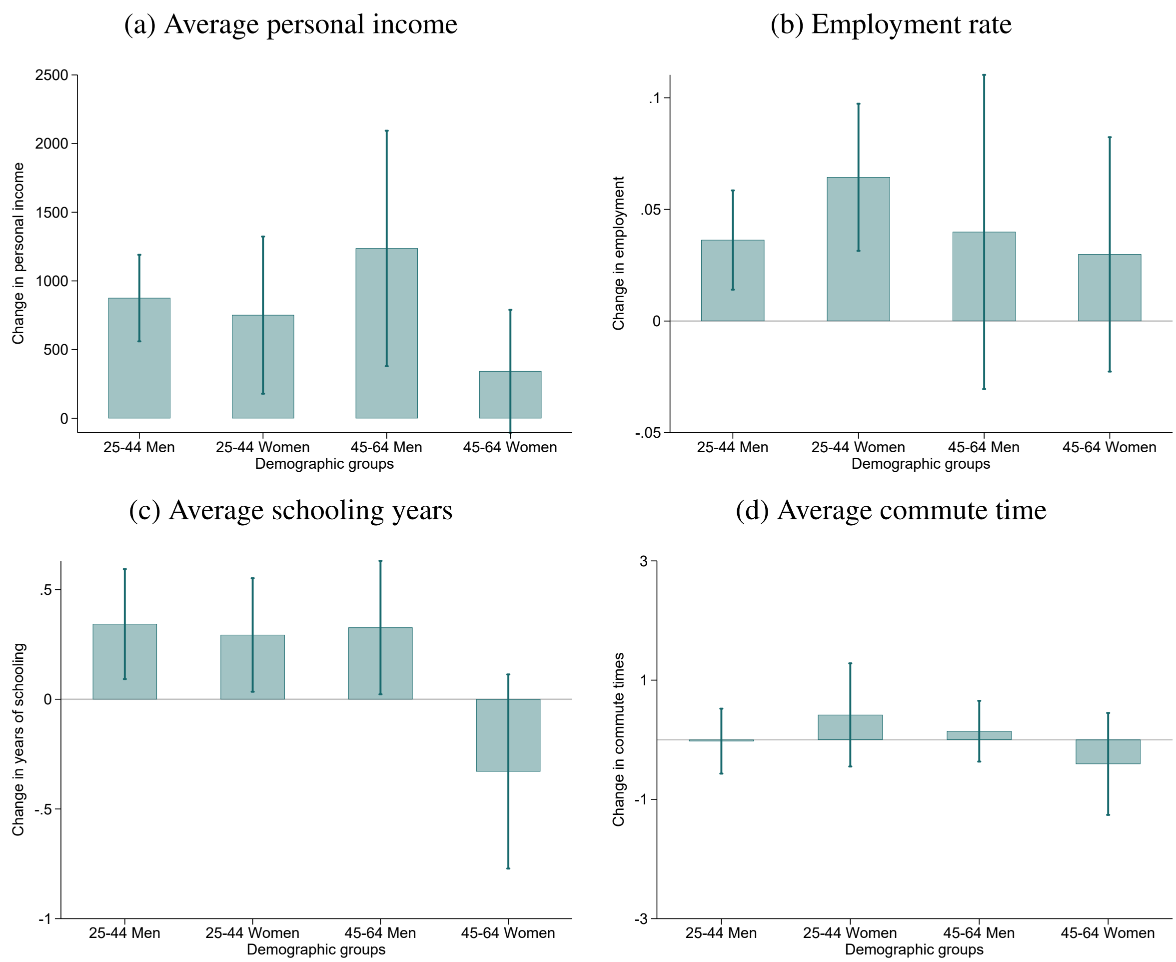
Figure 10 Panel (a) shows that, by Year 5, the average income of older men (aged 45-64) rose by $1078 (HKD), equaling with 16 percent of the pre-treatment average, while income of older women only rose by $272 (14 percent). The average income of younger adults increased by around $900, 9 percent and 20 percent for men and women respectively .

Panel (b) shows that the increases in average income of young adults coincided with increases in employment. All younger working-age adults experienced an increase in employment rate, 2 p.p (or 2 percent) for men and 6 p.p (or 12 percent) for women by Year 5. For other demographic groups, however, the employment rates remain unchanged.

Panel (c) shows that TPS induced a large increase in the average years of schooling among young adults, suggesting that the increases in income are partly attributable changes in human capital. By Year 5, the average schooling for younger adults increased by 0.38 years. Specifically, the share of younger adults who are college graduates and with higher degrees increased (see Online Appendix Figure A10). The average years of schooling for older adults remained unchanged. Since I focus on residents aged 25-44, the vast majority have already past their schooling age, so it is unlikely that they are induced to increase in their human capital invest-

[21] Here, children is defined as all offspring residing with the parent.

Figure 10: Effects of TPS on average personal income and employment, by demographic group



Notes: Figure plots the effect of TPS on average real personal income, employment rate, average year of schooling and commute times in the second Census following treatment relative to that of the last Census year before treatment. Commute time is defined as the minutes it takes for a person to drive from the estates to their workplace, as defined by Google's Distance Matrix API, with time set at 8:00 am August 25, 2020. Other notes follow Figure 7.

ment in response to the policy. These increases in average schooling most likely reflect the fact that younger residents with more schooling became more likely to stay in the estates, while younger residents with less schooling became more likely to move out.

Panel (d) shows that TPS did not detectably reduce the average commute times of working people in treated estates. This finding is notable, since previous studies have shown that public housing in Hong Kong, both rental and ownership, features significant spatial misallocation, as exhibited by larger commuting distances of its residents relative to private-sector counterparts (Lui and Suen 2011). Consistent with highly limited residential mobility both before and after the subsidized sale, the lack of response in commute times suggests that TPS did not meaningfully alter the degree of spatial misallocation between workers and jobs.

Figure 11 shows that the average housing expenditure—defined as the sum of monthly rental and mortgage payments—fell dramatically in the TPS estates relative to control. Average user cost fell by $327 (HKD), or 25 percent relative to 1996 levels, in treated estates by Year 0. The decline deepened and reached $640, or 49 percent lower by Year 15. These estimates imply that mortgage payments were lower than counterfactual rent payments immediately after the rollout of TPS and further diverged over time. Pressure to fulfill mortgage obligations thus cannot explain the increase in average income.

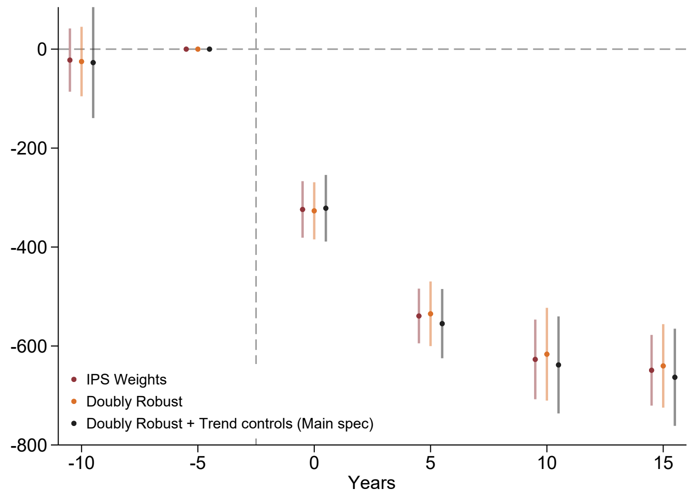
The above evidence suggests that within-person labor supply effects are unlikely to fully explain the observed changes in income distribution, since commute patterns were unchanged, housing expenditures fell, and changes in household composition appears to account for at least part of the increases in income.

5.4 Contribution to Subdivided Unit Crisis

This subsection illustrates how TPS may have affected the availability of public housing units to needy households. In the decade after TPS ended, Hong Kong gained international notoriety for the fact that its low-income populations became increasingly cramped into tiny subdivided units. These units were typically less than 15 square meter in size and have poor hygiene and safety conditions. According to the Population Census in 2016, there were 91,787 such units. The proliferation of these units has been universally condemned and is generally attributed to a shortage of public rental units. Between 2011 and 2019, as subsidized units proliferated, the average wait time for PRH applicants rose from 2.0 to 5.5 years.

The contribution of TPS to the subdivided unit crisis can be illustrated using back-of-the-envelope calculations leveraging the main estimates. Specifically, TPS caused two types of

Figure 11: Effects of TPS on average HH housing expenditure



Notes: Figure plots the effect of TPS on average real household housing expenditure. Online Appendix Table A5 displays coefficients and pre-treatment means. Other notes follow Figure 5.

units to become unavailable to low-income households who qualify for PRH units: the units that became unoccupied, and the units that became instead allocated to high-income households that do not qualify. A conservative estimate of this number (ignoring labor supply and general equilibrium price effects) can be formed by multiplying the number of units eligible for TPS purchases by the estimated effect of TPS on the share of households with incomes above two times the PRH income limit (i.e. households with 1.5$\times$ rent). The estimated number is $183700 \times 0.05 = 9185$. In other words, TPS can account for roughly $9185/91787 = 10\%$ of subdivided units in 2016.

6 Conclusion

This paper quantified the impacts of Hong Kong's Tenants Purchase Scheme—a natural experiment converting 183,700 subsidized rentals to ownership units—using its staggered rollout. A difference-in-differences design revealed that the program dramatically altered household incomes and sizes. Treated estates experienced a doubling in the share of households exceeding income eligibility thresholds. Concurrently, household size and population declined by 7-10 percent, the share of extended families shrank, even as average household incomes remained similar. Consistent with the fact that TPS did not confer purchasing households with the ability to transfer ownership or rent out units, there was very limited impact on household-level mobility. Moreover, average schooling increased, suggesting the changes in income are partly due to changes in demographic composition.

The evidence is strongly suggestive that extended family networks strategically reorganized co-residence choices in response to the scheme: higher-income family members stayed in the subsidized units, since they were no longer subject to regular income tests, and lower-income family members exited to receive additional public housing units. The resulting reduction in the availability of subsidized units for low-income populations was likely a contributor to the subsequent subdivided unit crisis. These findings underscore the importance of institutional details—such as income testing requirements and transfer restrictions—in mediating the impacts of housing assistance policy. In choosing whether to offer subsidized ownership or rental housing, careful consideration of institutional design is desired.

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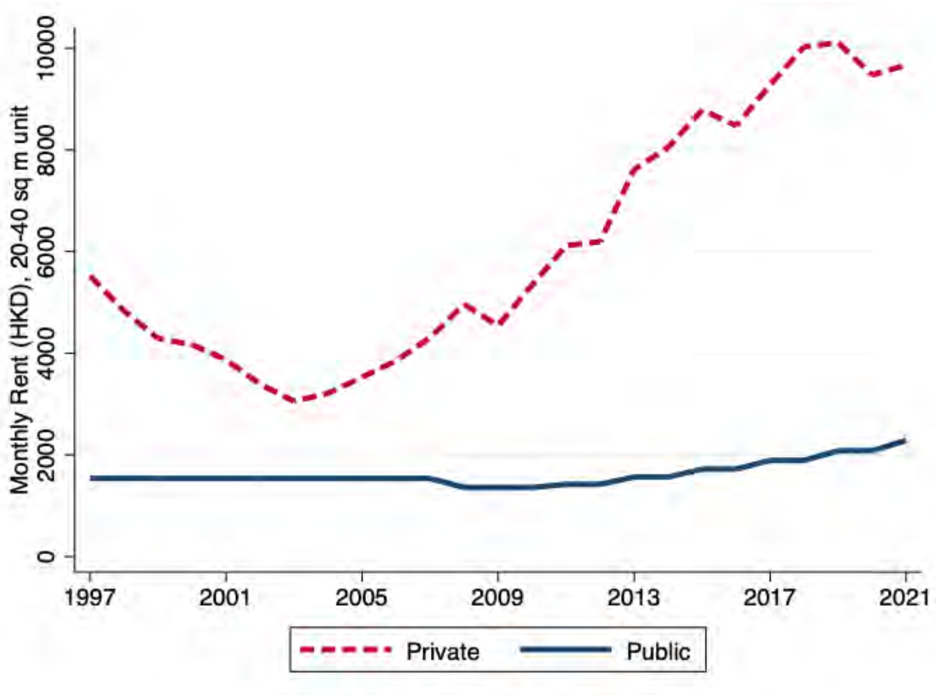
Online Appendix

A Additional Tables and Figures

Figure A1: Public and Private Rent for Similar Units

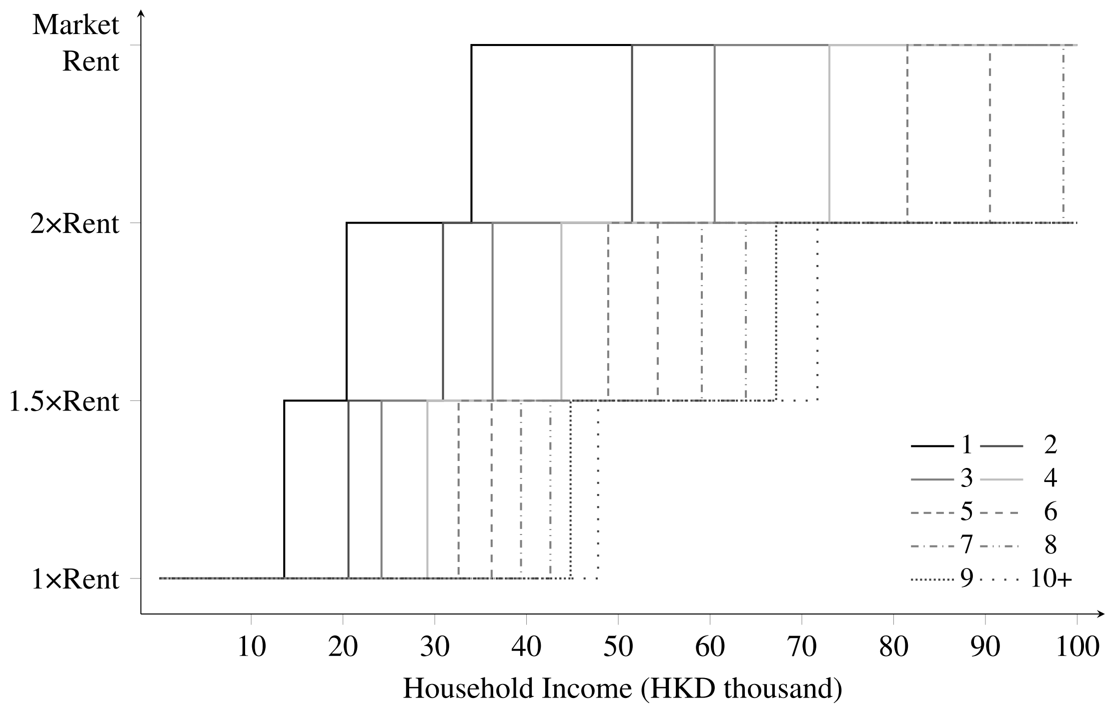


Notes: Figure plots rent indices for PRH and comparable private homes. The PRH rent index is constructed as follows. I first construct a PRH rent index with 2016 normalized to one using government announcements about the percentage changes in PRH rent. I then multiply the rent index by the average rent of households residing in 20-40 square meter PRH units in the 5% sample of the 2016 Hong Kong Population census. Note that 20-40 square meter units accounts for 67.2 percent of PRH housing stock in 2016. The rent index for comparable private sector homes is constructed as follows. First, I compute the average rent of comparable private homes in 2016. We take the average rent by district of renters in 20-40 square meter private-sector units in the 5% sample of the 2016 Hong Kong Population Census. I average across districts, with the number of 20-40 sq m PRH units in each district in 2016 as weights. Next, I obtain private-sector rent indices for Class A (i.e., <40 square meters) units by region (Hong Kong Island, Kowloon and New Territories) from the Rating and Valuation Department (RVD). I take the average across the RVD indices, weighted by the number of 20-40 square meter PRH units in each region. I then normalize 2016 to be the average rent of comparable private homes in 2016, as calculated from the Census data.

Table A1: Public rental housing (PRH) income limit in 2006

Household Size	**Income Limit (HKD)**
1 Person	6800
2 Person	10300
3 Person	12100
4 Person	14600
5 Person	16300
6 Person	18100
7 Person	19700
8 Person	21300
9 Person	22400
10 Person and above	23900

Figure A2: Rent schedule by household income and size



Notes: Under the "Well-off Tenants Policies", households with an income exceeding two times and not more than three times the prevailing PRH income limits have to pay 1.5 times net rent plus rates. Those with household income exceeding three times and not more than five times the prevailing PRH income limits are required to pay double net rent plus rates. PRH households with total household income or net assets value exceeding the prescribed limits (i.e. five times and 100 times of the PRH income limits respectively), as well as those who have private domestic property ownership in Hong Kong are required to vacate their PRH flats. These households have to secure accommodation on the private rental market at prevailing market rents. This figure illustrates the rent levels paid by households under the 2006 Public Rental Housing (PRH) Income Limits (see Appendix Table A1), as a function of household income and household size.

Table A2: Sample restrictions

	Treated	Control
All estates observed in Census years 1996-2016	39	97
No construction after 1996	39	72
No construction before 1980	39	43

Notes: Table counts the number of estates identified in the data and after imposing sample restrictions.

Table A3: List of estates

Treated estates, Cohort 1	Treated estates, Cohort 2	Control estates	
Cheung On Estate	Cheung Fat Estate	Ap Lei Chau Estate	Lower Wong Tai Sin (2) Estate
Choi Ha Estate	Cheung Wah Estate	Butterfly Estate	Lung Hang Estate
Chuk Yuen North Estate	Fu Shin Estate	Chak On Estate	Mei Lam Estate
Fu Heng Estate	Hing Tin Estate	Cheung Hang Estate	On Ting Estate
Fung Tak Estate	King Lam Estate	Choi Fai Estate	On Yam Estate
Fung Wah Estate	Kwai Hing Estate	Choi Yuen Estate	Sam Shing Estate
Heng On Estate	Kwong Yuen Estate	Chuk Yuen South Estate	Sha Kok Estate
Hin Keng Estate	Lei Cheng Uk Estate	Chun Shek Estate	Shek Wai Kok Estate
Kin Sang Estate	Lei Tung Estate	Hau Tak Estate	Shun Tin Estate
Tai Wo Estate	Leung King Estate	Hing Man Estate	Siu Sai Wan Estate
Tak Tin Estate	Long Ping Estate	Jat Min Chuen	Sun Chui Estate
Tin King Estate	Lower Wong Tai Sin (1) Estate	Ka Fuk Estate	Sun Tin Wai Estate
Tin Ping Estate	Nam Cheong Estate	Ka Wai Chuen	Tai Yuen Estate
Tsui Wan Estate	Po Lam Estate	Kai Yip Estate	Tin Shui (1) Estate
Wah Kwai Estate	Pok Hong Estate	Kwong Fuk Estate	Tin Shui (2) Estate
Wah Ming Estate	Shan King Estate	Kwong Tin Estate	Tin Yiu (1) Estate
Wan Tau Tong Estate	Tai Ping Estate	Kwun Tong Garden Estate	Tin Yiu (2) Estate
Yiu On Estate	Tsing Yi Estate	Lai Kok Estate	Tsz Man Estate
	Tsui Lam Estate	Lai On Estate	Wang Tau Hom Estate
	Tsui Ping North Estate	Lee On Estate	Wu King Estate
	Tung Tau (2) Estate	Lok Wah North Estate	Yiu Tung Estate
		Lok Wah South Estate	

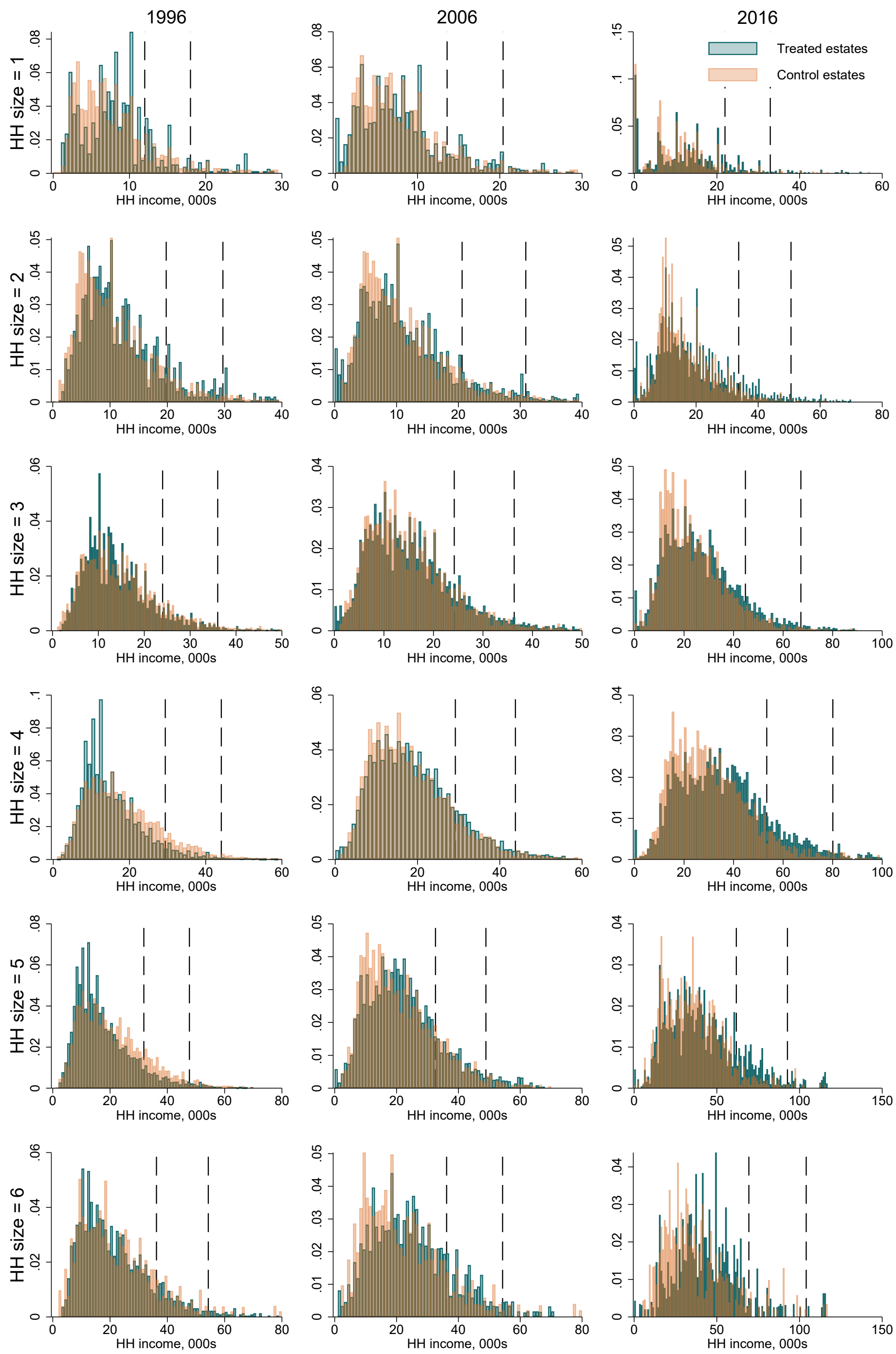
Notes: Table tabulates all estates included in analysis.

Table A4: HH characteristics, sold and unsold units in TPS estates, 2006

	Sold units	Unsold units	Standardized difference
HH size	3.52	2.91	0.45
	(1.3)	(1.36)	
HH income	18668	12853	0.49
	(13157)	(10304)	
Working persons per HH	1.84	1.24	0.54
	(1.16)	(1.09)	
HH with all 60+ y. o.	0.06	0.15	-0.29
	(0.24)	(0.36)	
Single-person	0.06	0.18	-0.36
Nuclear family	0.76	0.71	0.12
Extended family	0.38	0.32	0.17
Non-family	0.08	0.07	0.02
HH size = 1	0.06	0.18	-0.36
HH size = 2	0.16	0.23	-0.16
HH size = 3	0.25	0.25	0.01
HH size = 4	0.32	0.24	0.18
HH size = 5	0.15	0.08	0.23
HH size = 6+	0.06	0.03	0.11
Number of HHs	101112	80764	

Notes: Table shows mean household characteristics in TPS estates in 2006, respectively for TPS buyers and non-buyers.

Figure A3: HH income distribution by household size, treated vs control estates



Notes: Figure plots the distribution of household income in treated and control estates, respectively in 1996, 2006, and 2016. The 1.5X and 2X rent income limits are plotted in dashed vertical lines. Households with all members above age 60 are excluded.

Table A5: Effect of TPS on TPS take up, share of HH above cutoff, working person and housing cost

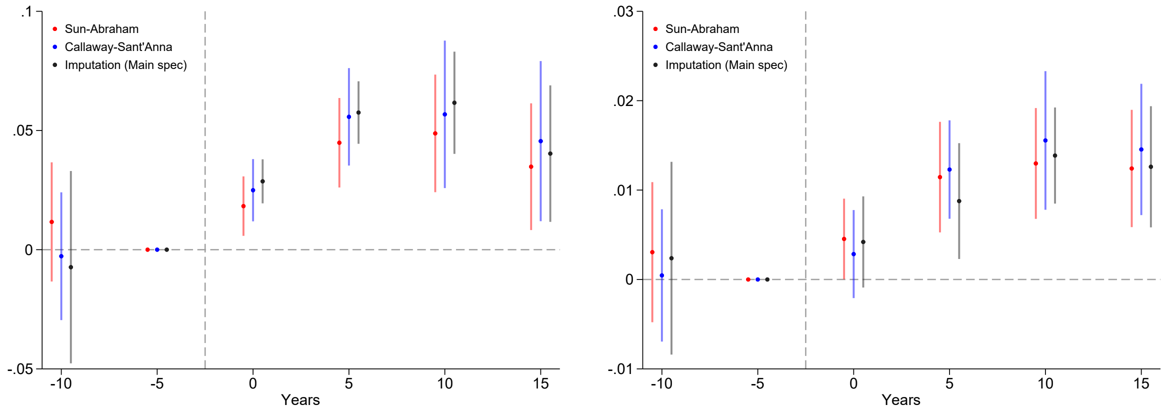
	(1) Share of TPS unites Sold	(2) Share of HH above 1.5X rent cutoff	(3) Share of HH above 2X rent cutoff	(4) Working persons per HH	(5) Housing cost per HH
Pre-trend					
t = -10	0.00013	-0.007	0.00238	-0.04	-27
	(0.00010)	(0.021)	(0.006)	(0.10)	(57)
Post					
t = 0	0.62**	0.029**	0.004	0.028	-322**
	(0.02)	(0.005)	(0.003)	(0.033)	(34)
t = 5	0.65**	0.058**	0.009**	0.070	-555**
	(0.02)	(0.007)	(0.003)	(0.060)	(36)
t = 10	0.71**	0.062**	0.014**	-0.016	-638**
	(0.02)	(0.011)	(0.003)	(0.091)	(50)
t = 15	0.77**	0.040**	0.013**	-0.100	-663**
	(0.02)	(0.015)	(0.003)	(0.102)	(50)
Treated mean, 1996	0	0.114	0.027	1.71	1296
Num. of estate-years	410	410	410	410	410
Num. of estates	82	82	82	82	82

Notes: Table reports coefficients estimated using the interaction-weighted specification of Borusyak, Jaravel and Spiess (2024), augmented with a doubly robust procedure, region-specific and construction-period-specific trends control. The sample includes all estates where all buildings were completed after 1979 and before 1996. "Year 0" denotes the first Census year observed after treatment. All reported treated means are IPS-weighted, in line with the regression weights. Standard errors, clustered at the estate level, are reported in parentheses. Significance levels: $\sim$ = 10%, * = 5%, ** = 1%.

Figure A4: Effects of TPS on rent15X and rent2X with alternative estimators

(a) Share of HHs above 1.5X rent cutoff

(b) Share of HHs above 2X rent cutoff

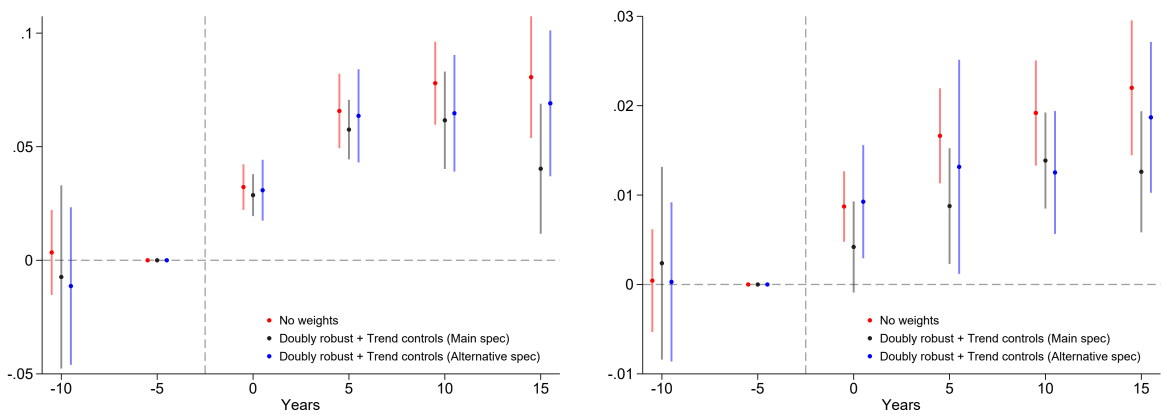


Notes: The black series plots coefficients from the interaction-weighted estimator of Borusyak, Jaravel and Spiess (2024), augmented with a doubly robust procedure, region-specific and construction-period-specific trends control. The red and blue series plots the same estimator, using Sun and Abraham (2020) and Callaway and Sant'Anna (2020).

Figure A5: Effects of TPS on rent15X and rent2X with alternative weights

(a) Share of HHs above 1.5X rent cutoff

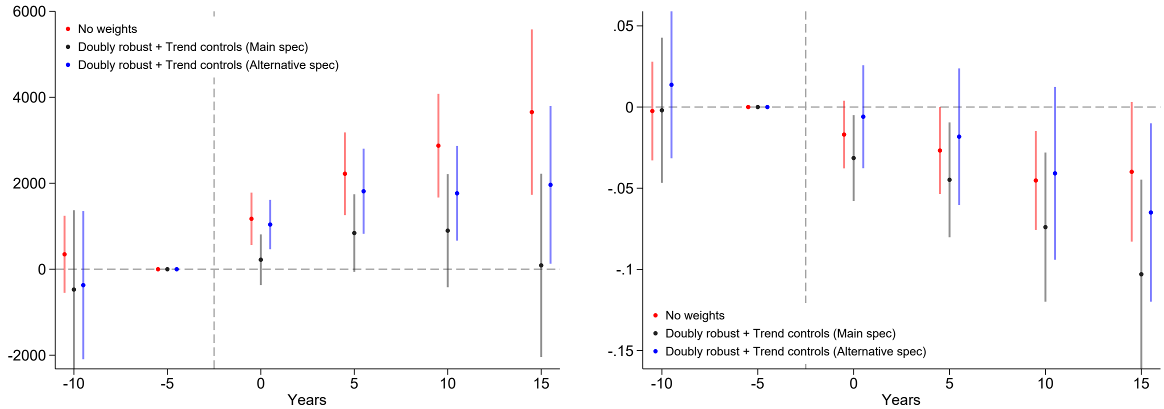
(b) Share of HHs above 2X rent cutoff



Notes: The black series corresponds to the main specification used in the paper, estimated using the interaction-weighted estimator of Borusyak, Jaravel and Spiess (2024) with doubly robust scores, region-specific and construction-period-specific trends control. The red series uses the same estimator without balancing weights. The blue applies the main specification but with alternative inverse propensity score weights based on year built and household-size composition.

Figure A6: Effects of TPS on HH incomes and HH size with alternative weights

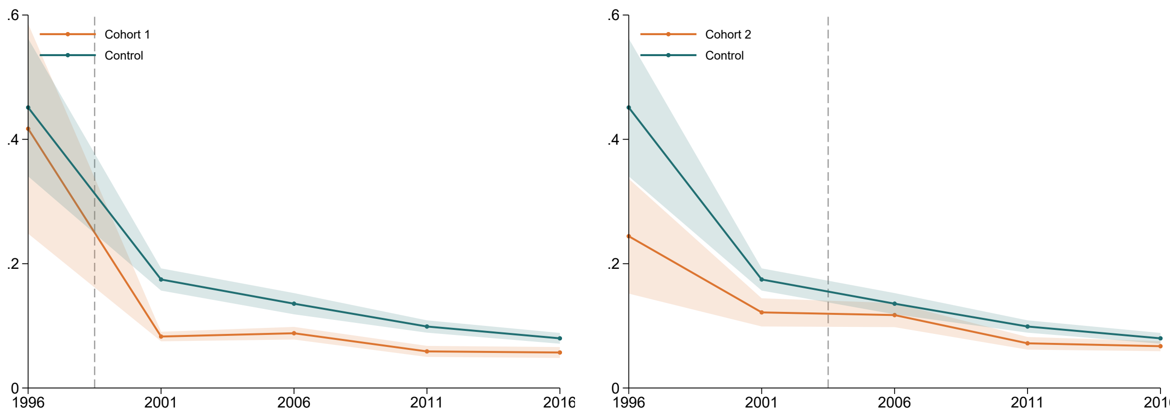
(a) Average HH Income (b) Log average HH size



Notes: See notes of Online Appendix Figure A5.

Figure A7: Trends in Share of HH moved in last 5 years, treated vs control estates

(a) Cohort 1 (b) Cohort 2



Notes: Each panel shows the trend in average share of household moved in last 5 years, separately for the two treated cohorts and controls. Cohort 1 includes estates that are treated before 2001, and Cohort 2 includes treated before 2006. Sample includes all estates where all buildings were built after 1979 and before 1996.

Table A6: Effect of TPS on HH size, population, number of HH and HH moved

	(1) Log average HH size	(2) Log population	(3) Log num. of HH	(4) Share of HH moved in last 5 years
Pre-trend				
t = -10	-0.002	-0.039	-0.037	0.03
	(0.023)	(0.050)	(0.034)	(0.08)
Post				
t = 0	-0.031*	-0.063**	-0.032**	-0.01
	(0.013)	(0.013)	(0.009)	(0.03)
t = 5	-0.045*	-0.057**	-0.012	0.0175
	(0.018)	(0.013)	(0.015)	(0.03)
t = 10	-0.074**	-0.097**	-0.023~	0.02
	(0.023)	(0.021)	(0.012)	(0.03)
t = 15	-0.103**	-0.129**	-0.026	0.00
	(0.030)	(0.029)	(0.018)	(0.04)
Treated mean, 1996	3.98	19430	4904	0.122
Num. of estate-years	410	410	410	410
Num. of estates	82	82	82	82

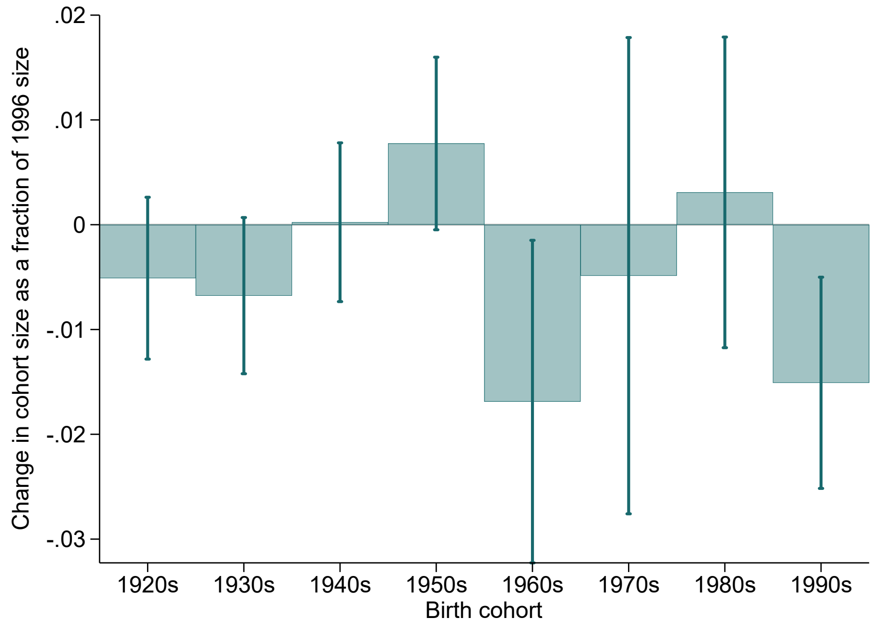
Notes: Treated mean shown for Log average HH size, Log population and Log num of HH is represented by average HH size, population and num of HH. In Column (4), unweighted mean in 2001 for Cohort 2 is displayed instead. Other notes follow Online Appendix Table A5

Table A7: Unit ownership of households in TPS estates over time

Year	1996	2001	2006	2011	2016
Share of HHs in unsold TPS units	100.0%	68.9%	42.6%	35.7%	28.1%
Share of HHs in sold TPS units	0.0%	31.1%	57.4%	64.3%	71.9%
TPS premium unpaid, Owner-occupied	0.0%	31.1%	55.6%	62.5%	70.9%
TPS premium unpaid, Rented	0.0%	0.0%	1.8%	1.4%	0.1%
TPS premium paid, Owner occupied	0.0%	0.0%	0.0%	0.3%	0.5%
TPS premium paid, Rented	0.0%	0.0%	0.0%	0.1%	0.4%
Number of households	185962	185641	181876	180022	177413

Notes: Table decomposes ownership status by household in TPS estates. Source: Hong Kong Population Census.

Figure A8: Effect of TPS on population by birth cohort

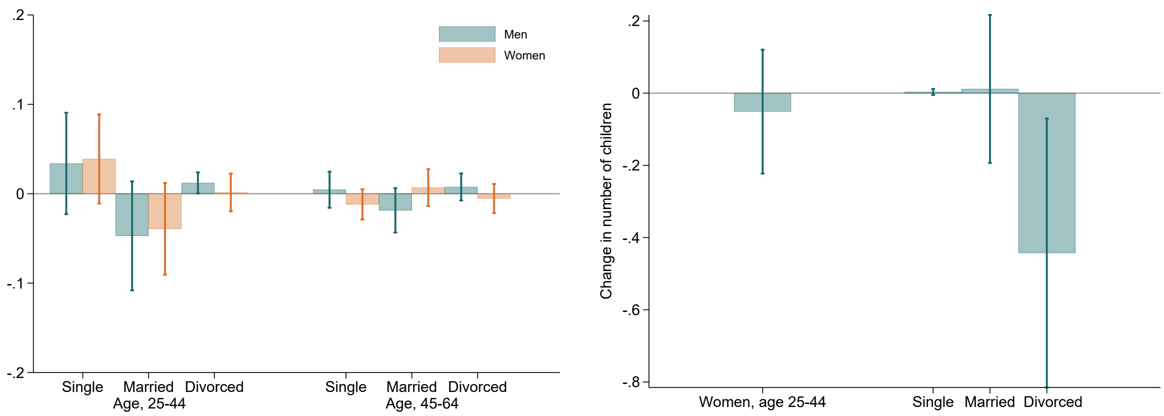


Notes: Figure plots the effect of TPS on the cohort size in treated estates as a fraction of 1996 cohort size in the second Census year following treatment relative to that of the last Census year before treatment, estimated using the interaction-weighted estimator in Borusyak, Jaravel and Spiess (2024), augmented with a doubly robust procedure, region-specific and construction-period-specific trends control. Other notes follow Figure 7.

Figure A9: Effects of TPS on share of married individual and number of children in treated estates

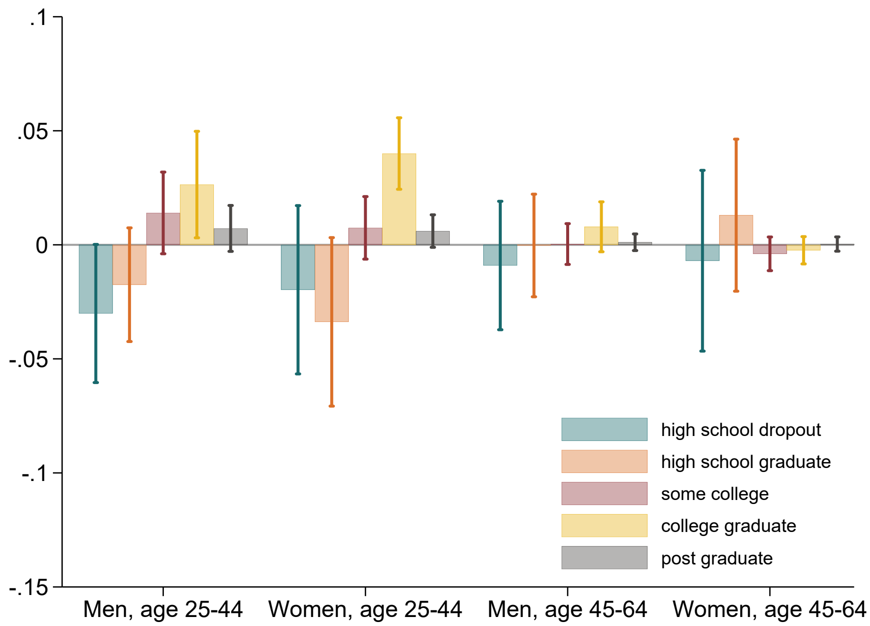
(a) Marital status

(b) Number of children



Notes: Panel A plots the effect of TPS on the share of single, married and divorced individual among different groups in the second Census following treatment relative to that of the last Census year before treatment, estimated using the interaction-weighted estimator in Borusyak, Jaravel and Spiess (2024), augmented with a doubly robust procedure, region-specific and construction-period-specific trends control. Panel B plots the effect of TPS on number of children among young women groups in different marital status in the second Census following treatment relative to that of the last Census year before treatment. Other notes follow Figure 7.

Figure A10: Effect of TPS on population by education attainment



Notes: Figure plots the effect of TPS on the share of education attainment in treated estates in the second Census year following treatment relative to that of the last Census year before treatment, estimated using the interaction-weighted estimator in Borusyak, Jaravel and Spiess (2024), augmented with a doubly robust procedure, region-specific and construction-period-specific trends control. Those with high school dropout refers to individuals who have not completed secondary education, or even lower. Those with postgraduate refers to individuals holding a master's degree or higher degree. Other notes follow Figure 7.

B Derivations: Model with Atomistic Households

B.1 Private sector

Households living in the private sector solve:

$$\begin{aligned} \max_{c,h,l} \quad & c^{\alpha} h^{\beta} l^{\gamma} \\ \text{s.t.} \quad & c + rh = w(T-l). \end{aligned}$$

Standard Lagrangian methods imply that optimal consumption, leisure, and housing are:

$$c_{Pri}(w) = \alpha w T; \quad l_{Pri}(w) = \gamma T; \quad h_{Pri}(w) = \frac{\beta}{r} w T.$$

The indirect utility of a household with wage w is

$$u_{Pri}(w) = \alpha^{\alpha} \beta^{\beta} \gamma^{\gamma} \frac{w^{\alpha+\beta}}{r^{\beta}} T,$$

and their income is

$$I_{Pri}(w) = (1-\gamma) w T.$$

B.2 Public Rental Housing

Households in public rental housing (PRH) units solve:

$$\begin{aligned} \max_{c,l} \quad & c^{\alpha} \overline{h}^{\beta} l^{\gamma} \\ \text{s.t.} \quad & c + R(w(T-l)) = w(T-l), \end{aligned}$$

where

$$R(I) = \begin{cases} \overline{R} & \text{if } I < 2\bar{I} \\ 1.5\overline{R} & \text{if } I \in [2\bar{I}, 3\bar{I}] \\ 2\overline{R} & \text{if } I > 3\bar{I}. \end{cases}$$

The above problem can be rewritten as follows:

$$\begin{aligned} \max_{c,l} \quad & c^{\alpha} \overline{h}^{\beta} l^{\gamma} \\ \text{s.t.} \quad & \begin{cases} c + \overline{R} = w(T-l) & \text{if } w(T-l) < 2\bar{I}, \\ c + \overline{R} = 2\bar{I} & \text{if } w(T-l) = 2\bar{I}, \\ c + 1.5\overline{R} = w(T-l) & \text{if } w(T-l) \in (2\bar{I}, 3\bar{I}), \\ c + 1.5\overline{R} = 3\bar{I} & \text{if } w(T-l) = 3\bar{I}, \\ c + 2\overline{R} = w(T-l) & \text{if } w(T-l) > 3\bar{I}. \end{cases} \end{aligned}$$

Consider each of the above five cases:

Case 1: $w(T-l) < 2\bar{I}$**.** Households solve:

$$\max_{c,l} \quad c^{\alpha}\overline{h}^{\beta}l^{\gamma}$$
$$\text{s.t.} \quad c+\overline{R} = w(T-l)$$

Optimal consumption and leisure in this case are given by

$$c_1(w) = \frac{\alpha}{\alpha+\gamma}\left(wT-\overline{R}\right); \quad l_1(w) = \frac{\gamma}{\alpha+\gamma}\left(T-\frac{\overline{R}}{w}\right).$$

The indirect utility is

$$u_1(w) = \alpha^{\alpha}\overline{h}^{\beta}\gamma^{\gamma}\frac{w^{\alpha}}{(\alpha+\gamma)^{\alpha+\gamma}}\left(T-\frac{\overline{R}}{w}\right)^{\alpha+\gamma},$$

and the income is

$$I_1(w) = \frac{\alpha}{\alpha+\gamma}wT+\frac{\gamma}{\alpha+\gamma}\overline{R}.$$

Case 2: $w(T-l) = 2\bar{I}$**.** The optimal consumption and leisure are:

$$c_2(w) = 2\bar{I}-\overline{R}; \quad l_2(w) = T-\frac{2\bar{I}}{w}.$$

Their indirect utility is

$$u_2(w) = \left(2\bar{I}-\overline{R}\right)^{\alpha}\overline{h}^{\beta}\left(T-\frac{2\bar{I}}{w}\right)^{\gamma}$$

and their income is $I_2(w) = 2\bar{I}$.

Case 3: $w(T-l) > 2\bar{I}$ **and** $w(T-l) < 3\bar{I}$**.** Households solve:

$$\max_{c,l} \quad c^{\alpha}\overline{h}^{\beta}l^{\gamma}$$
$$\text{s.t.} \quad c+1.5\overline{R} = w(T-l)$$

Optimal consumption and leisure are given by

$$c_3(w) = \frac{\alpha}{\alpha+\gamma}\left(wT-1.5\overline{R}\right); \quad l_3(w) = \frac{\gamma}{\alpha+\gamma}\left(T-\frac{1.5\overline{R}}{w}\right).$$

The indirect utility is

$$u_3(w) = \alpha^\alpha \overline{h}^\beta \gamma^\gamma \frac{w^\alpha}{(\alpha+\gamma)^{\alpha+\gamma}} \left(T - \frac{1.5\overline{R}}{w}\right)^{\alpha+\gamma},$$

and the income is

$$I_3(w) = \frac{\alpha}{\alpha+\gamma} wT + \frac{\gamma}{\alpha+\gamma} 1.5\overline{R}.$$

Case 4: $w(T-l) = 3\bar{I}$**.** The optimal consumption and leisure are:

$$c_4(w) = 3\bar{I} - 1.5\overline{R}; \quad l_4(w) = T - \frac{3\bar{I}}{w}.$$

The utility is

$$u_4(w) = \left(3\bar{I} - 1.5\overline{R}\right)^\alpha \overline{h}^\beta \left(T - \frac{3\bar{I}}{w}\right)^\gamma$$

and their income is

$$I_4(w) = 3\bar{I}.$$

Case 5: $w(T-l) > 3\bar{I}$**.**

Households solve:

$$\begin{aligned} \max_{c,l} \quad & c^\alpha \overline{h}^\beta l^\gamma \\ \text{s.t.} \quad & c + 2\overline{R} = w(T-l) \end{aligned}$$

The optimal consumption and leisure are given by

$$c_5(w) = \frac{\alpha}{\alpha+\gamma}\left(wT - 2\overline{R}\right); \quad l_5(w) = \frac{\gamma}{\alpha+\gamma}\left(T - \frac{2\overline{R}}{w}\right).$$

The indirect utility is

$$u_5(w) = \alpha^\alpha \overline{h}^\beta \gamma^\gamma \frac{w^\alpha}{(\alpha+\gamma)^{\alpha+\gamma}} \left(T - \frac{2\overline{R}}{w}\right)^{\alpha+\gamma},$$

and the income is

$$I_5(w) = \frac{\alpha}{\alpha+\gamma} wT + \frac{\gamma}{\alpha+\gamma} 2\overline{R}.$$

Summary

Note that $u_1(w) > u_3(w) > u_5(w)$. Moreover, Case 1 requires that:

$$w < \overline{w},$$

where $\overline{w} \equiv \frac{2(\alpha+\gamma)\bar{I}-\gamma\overline{R}}{\alpha T}$. Case 3 requires that

$$\underline{w} < w < 1.5\overline{w},$$

where $\underline{w} \equiv \frac{2(\alpha+\gamma)\bar{I}-1.5\gamma\overline{R}}{\alpha T}$. Case 5 requires that

$$w > \widetilde{w},$$

where $\widetilde{w} \equiv \frac{3(\alpha+\gamma)\bar{I}-2\gamma\overline{R}}{\alpha T}$. Note that $\underline{w} < \overline{w}$ and $\widetilde{w} < 1.5\overline{w}$.

Therefore, we have

$$u_{PRH}(w) = \begin{cases} \max\{u_1(w), u_2(w), u_4(w)\} & \text{if } w < \overline{w}, \\ \max\{u_2(w), u_3(w), u_4(w)\} & \text{if } \overline{w} \leq w < 1.5\overline{w}, \\ \max\{u_2(w), u_4(w), u_5(w)\} & \text{if } w \geq 1.5\overline{w}. \end{cases}$$

Income can be written as:

$$I_{PRH}(w) = \begin{cases} I_1(w) & \text{if } w < \overline{w} \text{ and } u_1(w) \geq \max\{u_2(w), u_4(w)\}, \\ I_2(w) & \text{if } w < \overline{w} \text{ and } u_2(w) \geq \max\{u_1(w), u_4(w)\}, \\ I_4(w) & \text{if } w < \overline{w} \text{ and } u_4(w) \geq \max\{u_1(w), u_2(w)\}, \\ I_2(w) & \text{if } w \in [\overline{w}, 1.5\overline{w}) \text{ and } u_2(w) \geq \max\{u_3(w), u_4(w)\} \\ I_3(w) & \text{if } w \in [\overline{w}, 1.5\overline{w}) \text{ and } u_3(w) \geq \max\{u_2(w), u_4(w)\}, \\ I_4(w) & \text{if } w \in [\overline{w}, 1.5\overline{w}) \text{ and } u_4(w) \geq \max\{u_2(w), u_3(w)\}, \\ I_2(w) & \text{if } w \geq 1.5\overline{w} \text{ and } u_2(w) \geq \max\{u_4(w), u_5(w)\}, \\ I_4(w) & \text{if } w \geq 1.5\overline{w} \text{ and } u_4(w) \geq \max\{u_2(w), u_5(w)\}, \\ I_5(w) & \text{if } w \geq 1.5\overline{w} \text{ and } u_5(w) \geq \max\{u_2(w), u_4(w)\}, \end{cases}$$

The chosen consumption and leisure can be derived in the same manner.

B.3 Tenants Purchase Scheme

Households living in Tenant Purchase Scheme (TPS) units solve:

$$\begin{aligned} \max_{c,l} \quad & c^{\alpha}\overline{h}^{\beta} l^{\gamma} \\ \text{s.t.} \quad & c + \overline{m} = w(T - l) \end{aligned}$$

Using standard Lagrangian methods, it can be found that optimal consumption, leisure, and housing are given by

$$c_{TPS}(w) = \frac{\alpha}{\alpha+\gamma}(wT - \overline{m}); \quad l_{TPS}(w) = \frac{\gamma}{\alpha+\gamma}\left(T - \frac{\overline{m}}{w}\right).$$

The indirect utility of a household with wage w is

$$u_{TPS}(w) = \alpha^{\alpha}\overline{h}^{\beta}\gamma^{\gamma}\frac{w^{\alpha}}{(\alpha+\gamma)^{\alpha+\gamma}}\left(T-\frac{\overline{m}}{w}\right)^{\alpha+\gamma},$$

and their income is

$$I_{TPS}(w) = \frac{\alpha}{\alpha+\gamma}wT + \frac{\gamma}{\alpha+\gamma}\overline{m}.$$

C Derivations: Model with Co-residence Choices

C.1 Public Rental Housing

First consider the joint utility when L and H live together in a public rental housing (PRH) unit. The household solves:

$$\begin{aligned} \max_{c_L,c_H} \quad & \frac{1}{\delta}\left(c_L^a\overline{h}^{1-a} + c_H^a\overline{h}^{1-a}\right) \\ \text{s.t.} \quad & c_L + c_H + R_2(I_L+I_H) = I_L + I_H. \end{aligned}$$

The optimal consumption of individual i is

$$c_{CoPRH} = \frac{1}{2}(I_L + I_H - R_2(I_L+I_H)),$$

and the indirect joint utility is

$$V_{CoPRH} = \frac{1}{\delta}\left(2\overline{h}\right)^{1-a}[I_L + I_H - R_2(I_L+I_H)]^a.$$

Second consider the utilities when individuals H and L live alone. There are three possible cases:

Case 1: H and L each lives in a PRH unit alone. Individual $i \in \{L,H\}$ solves:

$$\begin{aligned} \max_{c} \quad & c^a\overline{h}^{1-a} \\ \text{s.t.} \quad & c + R_1(I_i) = I_i. \end{aligned} \tag{2}$$

Individual i's optimal consumption is

$$c_{PRH}(I_i) = I_i - R_1(I_i),$$

and the indirect utility is

$$v_{PRH}(I_i) = (I_i - R_1(I_i))^a\overline{h}^{1-a}.$$

The joint utility is

$$V_{PRH,PRH} = (I_L - R_1(I_L))^a \overline{h}^{1-a} + (I_H - R_1(I_H))^a \overline{h}^{1-a}.$$

Case 2: L lives in a PRH unit and H lives in a private unit. L solves (2) and the indirect utility is

$$v_{PRH}(I_L) = (I_L - R_1(I_L))^a \overline{h}^{1-a}.$$

H solves:

$$\begin{aligned} \max_{c,h} \quad & c^a h^{1-a} \\ \text{s.t.} \quad & c + rh = I_H. \end{aligned} \tag{3}$$

H's optimal consumption is

$$c_{Pri}(I_H) = aI_H,$$

optimal housing is

$$h_{Pri}(I_H) = \frac{1-a}{r} I_H,$$

and indirect utility is

$$v_{Pri}(I_H) = \frac{a^a(1-a)^{1-a}}{r^{1-a}} I_H.$$

The joint utility is

$$V_{PRH,Pri} = (I_L - R_1(I_L))^a \overline{h}^{1-a} + \frac{a^a(1-a)^{1-a}}{r^{1-a}} I_H.$$

Case 3: L lives in a private unit and H lives in a PRH unit. L solves (3) (replacing I_H with I_L) and the indirect utility is

$$v_{Pri}(I_L) = \frac{a^a(1-a)^{1-a}}{r^{1-a}} I_L.$$

H solves (2) and the indirect utility is

$$v_{PRH}(I_H) = (I_H - R_1(I_H))^a \overline{h}^{1-a}.$$

The joint utility is

$$V_{Pri,PRH} = \frac{a^a(1-a)^{1-a}}{r^{1-a}} I_L + (I_H - R_1(I_H))^a \overline{h}^{1-a}.$$

Summary

Note that $V_{PRH,PRH} > V_{Pri,PRH}$ since we assume $v_{PRH}(I_L) > v_{Pri}(I_L)$. So H and L jointly maximize over $\{V_{CoPRH}, V_{PRH,PRH}, V_{PRH,Pri}\}$ and make co-residence choices accordingly.

C.2 Tenants Purchase Scheme

First consider the joint utility when they live together in a tenants purchase scheme (TPS) unit. The household solves:

$$\begin{aligned} \max_{c_L,c_H} \quad & \frac{1}{\delta}\left(c_L^a \overline{h}^{1-a} + c_H^a \overline{h}^{1-a}\right) \\ \text{s.t.} \quad & c_L + c_H + \overline{m} = I_L + I_H. \end{aligned}$$

The optimal consumption of individual i is

$$c_{CoTPS} = \frac{1}{2}(I_H + I_L - \overline{m}),$$

and the indirect joint utility is

$$V_{CoTPS} = \frac{1}{\delta}\left(2\overline{h}\right)^{1-a}(I_L + I_H - \overline{m}).$$

Second consider the utilities when individuals H and L live alone. There are four possible cases:

Case 1: L lives in a TPS unit and H lives in a PRH unit. L solves:

$$\begin{aligned} \max_{c} \quad & c^a \overline{h}^{1-a} \\ \text{s.t.} \quad & c + \overline{m} = I_L. \end{aligned} \tag{4}$$

L's optimal consumption is

$$c_{TPS}(I_L) = I_L - \overline{m},$$

and the indirect utility is

$$v_{TPS}(I_L) = (I_L - \overline{m})^a \overline{h}^{1-a}.$$

H solves (2) and the indirect utility is

$$v_{PRH}(I_H) = (I_H - R_1(I_H))^a \overline{h}^{1-a}.$$

The joint utility is

$$V_{TPS,PRH} = (I_L - \overline{m})^a \overline{h}^{1-a} + (I_H - R_1(I_H))^a \overline{h}^{1-a}.$$

Case 2: L lives in a TPS unit and H lives in a private unit. L solves (4) and the indirect utility is

$$v_{TPS}(I_L) = (I_L - \overline{m})^a \overline{h}^{1-a}.$$

H solves (3) and the indirect utility is

$$v_{Pri}(I_H) = \frac{a^a(1-a)^{1-a}}{r^{1-a}} I_H.$$

The joint utility is

$$V_{TPS,Pri} = (I_L - \overline{m})^a \overline{h}^{1-a} + \frac{a^a(1-a)^{1-a}}{r^{1-a}} I_H.$$

Case 3: L lives in a PRH unit and H lives in a TPS unit. L solves (2) and the indirect utility is

$$v_{PRH}(I_L) = (I_L - R_1(I_L))^a \overline{h}^{1-a}.$$

H solves (4) (replacing I_L with I_H) and the indirect utility is

$$v_{TPS}(I_H) = (I_H - \overline{m})^a \overline{h}^{1-a}.$$

The joint utility is

$$V_{PRH,TPS} = (I_L - R_1(I_L))^a \overline{h}^{1-a} + (I_H - \overline{m})^a \overline{h}^{1-a}.$$

Case 4: L lives in a private unit and H lives in a TPS unit. L solves (3) (replacing I_H with I_L) and the indirect utility is

$$v_{Pri}(I_L) = \frac{a^a(1-a)^{1-a}}{r^{1-a}} I_L.$$

H solves (4) (replacing I_L with I_H) and the indirect utility is

$$v_{TPS}(I_H) = (I_H - \overline{m})^a \overline{h}^{1-a}.$$

The joint utility is

$$V_{Pri,TPS} = \frac{a^a(1-a)^{1-a}}{r^{1-a}} I_L + (I_H - \overline{m})^a \overline{h}^{1-a}.$$

Summary

Note that $V_{PRH,TPS} > V_{Pri,TPS}$ since we assume $v_{PRH}(I_L) > v_{Pri}(I_L)$. So H and L jointly maximize over $\{V_{CoTPS}, V_{TPS,PRH}, V_{TPS,Pri}, V_{PRH,TPS}\}$ and make co-residence choices accordingly.